Explainable artificial intelligence for neural biomarker identification in Parkinson's disease

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September 2026

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WORD COUNT

Number of Pages: 70

Number of Words: 19711

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Dissertation Title

Explainable artificial intelligence for neural biomarker identification in Parkinson's disease

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Abstract

Parkinson's disease is treated with levodopa, a drug that relieves motor symptoms for a limited period after each dose. As the disease advances, patients begin to fluctuate between an ON state, in which the medication is working, and an OFF state, in which its effect has worn away. Identifying which state a patient occupies, continuously and objectively rather than by clinical impression, would allow treatment to be matched to the patient's actual condition. The established physiological marker of the untreated state is beta-band activity recorded from electrodes implanted in the brain, which is available only to the minority of patients carrying suitable hardware. This dissertation asked whether the same information survives at the scalp, in electroencephalography (EEG) recorded with a dry-electrode headset while patients moved freely through everyday activities, and which frequencies carry it.

Recordings from 35 people with Parkinson's disease were analysed. Each attended a single visit to an apartment-laboratory and completed the same household tasks twice, once after levodopa had taken effect and once after it had worn off. A large, pre-trained EEG model, CodeBrain, was adapted by three methods of increasing flexibility, a frozen probe, low-rank adaptation of 0.6% of the model, and full fine-tuning of all 15.2 million parameters. The main analysis trained one shared model on 10 second windows from one kitchen task and tested it on another, so that any solution had to survive a change of behavioural context. Estimates were checked against controls built from features that cannot carry brain activity, and thresholds were fixed before results were examined.

Medication state was separable within participants across a change of behavioural task. Full fine-tuning reached a participant-mean balanced accuracy of 0.832 (95% CI 0.792–0.869) in the principal direction, against 0.525 for a control built on per-channel amplitude alone, with accuracy rising monotonically with adaptation capacity. The separation did not survive removal of the test participant's own recordings from training, where every adapted configuration returned to chance and a pairwise test across 702 participant pairs found none that transferred. Classical relative band power was the only feature family to cross between individuals, weakly, at 0.620; features confined to the beta band performed at chance. A band-stop sweep showed the model reads a distributed amplitude-spectrum signature rather than a localised waveform event, with a single located dependence between 4 and 10 Hz surviving a control for the quantity of power removed, in one direction of the split only. No candidate met the certification criterion fixed in advance. The contribution is therefore a boundary on what ambulatory dry-electrode EEG supports, not a biomarker.

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List of abbreviations

Abbreviation	Definition
AI	Artificial intelligence
AUC	Area under the curve
BA	Balanced accuracy
CI	Confidence interval
DBS	Deep brain stimulation
DSI-24	Dry-electrode EEG headset used for recording (Wearable Sensing DSI-24)
EEG	Electroencephalography
ICA	Independent component analysis
LFP	Local field potential
LOPO	Leave-one-participant-out (validation design)
LoRA	Low-rank adaptation
ML	Machine learning
PD	Parkinson's disease
SNR	Signal-to-noise ratio
STN	Subthalamic nucleus
UPDRS	Unified Parkinson's Disease Rating Scale
XAI	Explainable artificial intelligence

Acknowledgements

I would like to thank my supervisor, Dr Shlomi Haar, for his guidance, feedback and support throughout this project.

I would also like to acknowledge the authors of CodeBrain (Ma et al., 2025) for making their pretrained EEG foundation model publicly available, which made the adaptation and interpretability work in this dissertation possible.

Artificial intelligence tools were used during the preparation of this dissertation to assist with tasks such as editing, formatting and organising material. Although all analysis, results, interpretation and writing represent my own work.

1. Introduction

1.1 Parkinson’s disease and its treatment

Parkinson's disease (PD) is the second most common neurodegenerative disorder worldwide, and it is becoming more common. Prevalence rose from roughly 2.5 million cases in 1990 to over 8.5 million by 2019 (Ou et al., 2021), reaching an estimated 11 million by 2021 (Luo et al., 2025). Most of that growth reflects an ageing population rather than any change in risk. Clinically, PD is defined by a cardinal motor triad of resting tremor, rigidity and bradykinesia. It also produces a wide range of non-motor symptoms, and these predict health-related quality of life more reliably than motor severity does (Kalia and Lang, 2015; Postuma et al., 2015; Martinez-Martin et al., 2011). The disease is also markedly heterogeneous, two patients with the same diagnosis may differ in how fast they decline, which symptoms dominate and how well they respond to the same drug. Nothing available halts the underlying neurodegeneration. Treatment is therefore symptomatic only, and for more than half a century it has centred on levodopa, a dopamine precursor able to cross the blood-brain barrier (Pedrosa de Menezes et al., 2024; Kalia and Lang, 2015; Poewe et al., 2017). As the disease progresses and the capacity to store dopamine declines, the window of benefit from each dose narrows. Patients then begin to fluctuate between an ON state, in which the medication is working and movement is close to normal, and an OFF state, in which its effect has worn away and symptoms return (Poewe et al., 2017). Determining which of these two states a patient occupies, objectively and continuously rather than by clinical impression, is the problem this dissertation addresses.

1.2 The neural basis of Parkinson’s disease

The motor features of PD arise from the progressive loss of dopaminergic neurons in the substantia nigra pars compacta, whose axons project to the striatum and supply the dopamine that regulates activity throughout the basal ganglia (Kalia and Lang, 2015). The basal ganglia are a group of subcortical nuclei, principally the striatum, the globus pallidus (GPi) (external and internal segments), the subthalamic nucleus (STN) and the substantia nigra. Together these form part of a set of re-entrant loops that run from the cortex, through these nuclei and the thalamus, and back to the cortex. Within this circuit, dopamine acts differentially on two opposing pathways, it facilitates transmission along the direct pathway, which disinhibits the thalamus and so promotes movement, while it suppresses transmission along the indirect pathway, the route that passes through the STN and inhibits the thalamus (Albin et al., 1989). Striatal dopamine depletion shifts the balance toward the indirect pathway. Inhibitory output from the internal globus pallidus to the thalamus rises, thalamocortical drive to the motor cortex drops, and the result is the slowness and poverty of movement that characterise the disease (DeLong, 1990; Kalia and Lang, 2015; Poewe et al., 2017). Symptoms typically only emerge once roughly half of the nigral population has degenerated, because surviving neurons compensate for those already lost. This is one reason PD is difficult to identify early (Poewe et al., 2017). Degeneration is not confined to the nigrostriatal system either. α-synuclein pathology spreads across brainstem, limbic and cortical regions, and this distribution accounts for the breadth of the non-motor presentation (Poewe et al., 2017).

This account in terms of firing rates alone is incomplete, though. Dopamine depletion also alters the temporal pattern of activity within the cortico-basal ganglia loop, and it is this change in patterning, rather than any change in rate, that produces the electrophysiological signatures the biomarker literature, and this dissertation, depend on.

1.3 Electrodes to Electroencephalogram (EEG)

Reliably identifying whether a patient is in the ON or OFF medication state would allow clinicians to adjust dosing more precisely and let researchers study symptom fluctuations directly. Doing so requires a measurement of what the brain is doing, and two recording modalities recur throughout this dissertation. The first is the local field potential (LFP), introduced here first because it is the current gold standard against which other modalities, including EEG, are typically compared, a recording of the electrical activity in a small region of brain tissue, taken directly from the Deep Brain Stimulous (DBS) electrodes themselves. The sensor sits inside the brain, so LFP signals are strong, clean, and rich in information. They require a sensing-capable DBS device, and not every implanted device has that capability, and they sample only the small volume of tissue immediately around the electrodes, which means they carry no information about activity elsewhere in the motor network. The cohort analysed in this dissertation carries no implanted device, so LFP is cited throughout for the physiology it established rather than as a modality available here; where the invasive literature is drawn on in Chapter 2, it is for the evidence it provides about the signals themselves and not for the therapy that produced it.

Where LFP requires an implanted, sensing-capable device and captures only a small volume of tissue, the second modality, electroencephalography (EEG), offers a non-invasive alternative with much broader coverage. EEG is a much older and much cheaper technique in which small electrodes are placed on the scalp to record the summed electrical activity of large populations of cortical neurons. EEG is entirely non-invasive, widely available, and covers the whole brain, but the signal it picks up is weaker than an LFP and heavily contaminated by muscle activity, eye movements, and interference from any electrical devices nearby, including the DBS device itself. Roy et al. (2019), in a systematic review of deep-learning approaches to EEG, catalogue both the promise of the method and its recurring failure modes. The signature that has dominated this literature is beta-band activity. Recording LFP directly from implanted electrodes, Brown et al. (2001) showed that oscillatory synchrony between the STN and the pallidum below roughly 30 Hz is prominent in the untreated state and suppressed by dopaminergic medication, establishing a direct link between the dopamine depletion described above and a measurable feature of the signal. Subsequent work established that the power of this beta rhythm scales with the severity of bradykinesia and rigidity, and that it falls as symptoms improve under levodopa (Kühn et al., 2006, 2009). Exaggerated beta synchrony is consequently the most extensively validated electrophysiological marker of the dopamine-depleted state, and it is the reference point against which any alternative candidate must be judged. Chapter 2 examines that evidence base, and its limits, in detail. This signature was established invasively, from an electrode sitting within the STN itself. At the scalp, the signal is spatially smeared, attenuated by the skull, and summed with activity from the whole cortex. Whether an equivalent, medication-

sensitive signature survives that process is a separate question, one this dissertation tests directly.

1.4 Where machine learning fits in

Recording brain activity produces enormous quantities of data. ML is a set of methods that allow a computer to learn patterns directly from data, rather than being explicitly programmed with fixed rules for every case. Deep learning is a subset of ML that uses artificial neural networks with many layers, and it has driven most of the major advances in image and speech processing over the last decade (LeCun et al., 2015). Applied to EEG, it offers a route from classification accuracy to a description of the cortical patterns the model has learned to recognise. This is the central aim of the present project, not simply to classify medication state as accurately as possible, but to use Explainable AI (XAI) to surface interpretable, physiologically meaningful candidate biomarkers from the EEG signal.

It also brings new problems, networks trained on small datasets can fail to generalise to patients they were not trained on, and their outputs can be difficult to interpret. Much of Chapter 2 is concerned with when and how these methods succeed or fail when applied to Parkinsonian brain signals. The interpretability problem is not a side issue in a clinical setting. A network that predicts medication state accurately but cannot say what in the signal it is responding to gives a clinician nothing to act on and gives a researcher no new physiological insight. This trade-off is a recurring one in deep learning, the more complex a model becomes, the harder it typically is to interpret why it produced a given output. XAI refers to a set of techniques for extracting that information. Asking a trained model which features of the input drove its decision.

1.5 Objectives and structure of this dissertation

The remainder of this dissertation moves from a broad survey of the field to a specific technical contribution. Chapter 2 reviews the literature on biomarkers for PD and DBS, what has been tried, what works, and where the gaps sit. Chapters 3 and 4 describe the methods and results of this project, which applies explainable AI to a deep-learning model of scalp EEG in order to characterise cortical patterns associated with PD medication state, not to replace existing biomarkers such as beta, but to ask what else the signal contains that a method unconstrained by any single hand-selected feature might find. Chapter 5 discusses the implications of the findings and locates them within the wider effort to make PD treatment more responsive to the individual patient, before a final chapter draws the dissertation's conclusions. The intended reader is someone with a scientific background but not necessarily a specialist in movement disorders or neural signal processing. The literature review that follows introduces technical detail incrementally, but the material in this introduction is intended to be enough to follow the argument even without prior exposure to the field.

The aim of this dissertation is to determine whether scalp EEG recorded outside the clinic carries enough information to distinguish a patient's ON and OFF levodopa state, and, where such information is present, to characterise the frequency content that carries it. Four objectives follow. The first is to adapt a large, pre-trained EEG foundation model to this task through parameter-efficient fine-tuning, so that representations learned from a large corpus of clinical recordings can be brought to bear on a comparatively small patient cohort.

The second, which is the main analysis of the dissertation, is to evaluate that model under a pooled task-split design, a single shared model is trained on windows from one kitchen task and tested on windows from another, so that generalisation is required across behavioural context, and both directions of the split are reported. The third is to apply post-hoc perturbation-based explainability, in the form of a band-stop frequency sweep, to the arm that scores highest under that design, to establish which frequency bands its decisions depend upon. The fourth is to establish the boundary of the result by re-running the same protocol with the test patient's recordings withheld from training, and by comparing throughout against a classical spectral baseline and against controls built on features that cannot carry cortical information. An earlier leave-one-patient-out design, evaluated on a wider cohort before the task split was adopted, is reported as exploratory work. The emphasis falls throughout on the third objective, classification accuracy is treated as a precondition for interpretation rather than as the result of interest.

2. Literature review

2.1 The clinical problem

Parkinson's disease is biologically and clinically heterogeneous, with subtypes progressing at markedly different rates and responding unevenly to available therapies (De Pablo-Fernández et al., 2019; Dadu et al., 2022). Because no two patients respond to stimulation in the same way, there is no standardised parameter setting that can be applied across patients, clinicians must instead search for the right combination individually for each person. For patients on DBS, that heterogeneity is therefore embedded directly in the programming process itself, which remains iterative and subjective (Duffley et al., 2023; Louie et al., 2021), guided by clinical rating scales. Principally, the Unified Parkison’s Disease Rating Scale (UPDRS), which is reliable at the summary level when administered by trained specialists (Siderowf et al., 2002), cannot resolve changes below a minimal clinically important difference of roughly 3–5 points (Horváth et al., 2015). Small parameter adjustments therefore produce physiological effects below the resolution of the instrument used to select them, so a clinician cannot reliably tell whether an adjustment has helped. Settings are then held fixed between visits while the underlying neural state fluctuates substantially. These limitations motivate the search for an objective physiological biomarker, one that resolves finer change than a rating scale and could, in principle, support adaptive stimulation.

2.2 Beta, the incumbent biomarker

The exaggerated 13–30 Hz synchrony recorded from the STN is an extensively documented electrophysiological signature of PD. This oscillatory synchrony is suppressed by dopaminergic medication (Brown et al., 2001), and subsequent work established that its spectral power (the strength of the oscillation) scales with bradykinesia and rigidity (Kühn et

al., 2009) and its power decreases alongside symptom improvement under both levodopa and high-frequency stimulation (Kühn et al., 2006, 2008).

In the largest cohort to test this relationship, Neumann et al. (2016) reported a moderate correlation ($\rho = 0.44$) between subthalamic 8–35 Hz activity and UPDRS-III scores. Later analyses refined this understanding by showing that beta is not tonic but bursty, and that the duration and amplitude of individual bursts track symptom severity and are shortened by adaptive stimulation (Tinkhauser et al., 2017a, 2017b; Lofredi et al., 2023). Beta-driven adaptive DBS has now met its primary endpoint in an industry-sponsored pivotal trial (Bronte-Stewart et al., 2025) and received regulatory approval. Beta is therefore already a deployed therapy, not merely a hypothesis, and any candidate biomarker proposed to displace or supplement it must be justified in that light.

The problem is not that beta is wrong, but that its coverage is quantifiably incomplete, for two distinct reasons, a biological limit on how many patients carry a detectable beta signal in the first place, and a practical, technology-dependent limit on how reliably that signal can be found even when it is present.

The biological limit is anatomical, adaptive stimulation is delivered to each hemisphere independently, so a usable signal is required on both sides, not just in the patient overall. Darcy et al. (2022) found detectable beta peaks in 92% of patients who were off medication, the condition in which beta is strongest and detection should be easiest, but in only 84% of individual hemispheres. Stanslaski et al. (2024), screening patients for the ADAPT-PD adaptive stimulation trial, found the same pattern under more clinically realistic conditions, on medication, the state in which patients live, clinicians identified a usable signal in 84.8% of patients but in both hemispheres in only 65%. Detection also varied by target, falling to 78% in the STN against 100% in the GPi, and the authors caution that even these figures may be optimistic, since some centres pre-screened participants for peak presence before enrolment. The practical limit is technological rather than biological. An offline automated algorithm, evaluated on the same Stanslaski et al. (2024) patients, found peaks in only 71.4% of cases. A gap between what an expert reader and a deployed detection algorithm can find, and one that matters because a real-world adaptive system must locate the signal without an expert reading it.

Real-world data suggest a lower ceiling still. Cascino et al. (2026), reporting on the ADAPT-START cohort of consecutive clinical patients rather than trial candidates, offered dual-threshold adaptive stimulation to 20 patients with sensing-capable devices, of whom nine proved eligible. The exclusions were driven by cardiac artefact and sensing-incompatible stimulation configurations rather than by absent beta activity. Technical constraints, restrict applicability to roughly half of assessed patients, a narrower window than the trial-cohort figures alone would suggest.

Even where beta is present, it is not stable. Van Rheede et al. (2022) documented that time of day alone accounted for approximately 41% of the variance in subthalamic beta power, and Behnke et al. (2025) subsequently showed that beta peak frequencies drift substantially across the first post-operative months, undermining the assumption that a threshold set at one clinic visit remains valid at the next. Bronte-Stewart et al. (2025) report improved outcomes for beta-driven DBS. Those outcomes, however, were established in patients whose signal

was present, in configurations capable of sensing it, and over the period across which the threshold was set. The trial makes no claim beyond those conditions.

These are limitations of a specific feature rather than of the enterprise, a narrowband oscillation that became the field's focus because it was the first strong candidate identified, and that has not been shown to be the most informative signal available. Every constraint named above follows from continuing to look for that same feature, and the question applies equally to patients in whom beta is perfectly measurable. Whether cortical activity beyond the beta band, recorded non-invasively and analysed without prior commitment to any single frequency, carries information that beta misses has not been well explored, and answering it requires a method that does not decide in advance which feature to look for.

2.3 Gamma and the artefact debate

Where beta has been most systematically studied at low frequencies, mid-gamma-range activity, and specifically the finely tuned gamma component sensitive to DBS, has become the most active alternative candidate, and the most contested. The central dispute concerns whether mid-gamma sensitivity to stimulation reflects a genuine neural response or a 1:2 subharmonic entrainment of the stimulation frequency itself, which would make it an artefact of the device rather than a physiological signal. This distinction matters directly for its use as a biomarker, a genuine physiological response could track disease or medication state independently of the stimulator, whereas an artefact would simply mirror the device's own output and carry no real information about the patient, making it useless or actively misleading as a control signal.

Sermon et al. (2023, 2024) made a detailed case for the former interpretation on the basis of modelling and empirical validation. They observed that stimulation-entrained gamma appears only in patients with pre-existing finely tuned gamma, and that it disappears above a critical stimulation amplitude, consistent with Arnold-tongue predictions from dynamical-systems theory. It can also be recorded at contacts remote from the stimulating electrode. Together, these features distinguish a genuine oscillatory response from a volume-conducted artefact. Mathiopoulou et al. (2025), extending this line of research, argued that entrained gamma predicts faster movement in patients and may itself serve as a closed-loop biomarker for stimulation control. In contrast, Wiest et al. (2021) reported that finely tuned gamma can be induced off medication without tracking the stimulation frequency and can outlast the stimulation period itself, both properties that are difficult to reconcile with simple entrainment. Entrainment, in their reading, is real but incomplete, it does not explain the full phenomenology on its own.

The most defensible position and the one this review adopts is that both describe part of the picture. Sermon et al.'s (2024) title, "1:2 entrainment is not a device-induced artefact, except when it is," captures the necessary nuance directly. Calvo Peiro et al. (2025), applying deep learning to scalp EEG during DBS parameter changes, found clear subharmonic entrainment in only 5 of 22 hemispheres. The classifier's accuracy dropped in roughly 3 of those 5 cases, suggesting that entrainment accounts for part, but not all, of the effect it was detecting. Their conclusion, that cortical mid-gamma involvement extends beyond 1:2 entrainment alone,

supports Wiest et al.'s position while acknowledging Sermon et al.'s mechanism where it applies.

This is the productive shape of the debate, neither side wins outright, instead, a joint account holds, in which an artefact-like mechanism operates in some patients, and a physiological signal operates in others, and the practical question becomes how to tell them apart in an individual case.

The objection that these signals are artefact rather than physiology has been met by direct disconfirmation rather than by assertion. Contact-change discrimination with matched stimulation amplitudes, bipolar-versus-monopolar comparisons, and analyses of entrainment-independent gamma each test the artefact hypothesis on its own terms (Sermon et al., 2023, 2024; Calvo Peiro et al., 2025). This dissertation applies the same scrutiny in reverse, rather than testing a single named candidate for artefact, it asks whether a data-driven approach, applied without commitment to any one feature, surfaces patterns that survive equivalent interrogation.

2.4 Recording modality, scalp EEG under stimulation

Having established which signals show promise as biomarkers, it is equally important to consider how they are measured, since a candidate's practical value depends on the accessibility of the recording modality as much as on its underlying physiology. Every biomarker discussed so far has been established primarily on invasive LFP recordings, the highest-fidelity source in this literature and the modality in which almost all clinical validation has been done. LFP is not, however, universally available, sensing-capable pulse generators, perioperative access to the electrode leads, or externalised electrodes during the immediate post-operative period are all preconditions that a significant fraction of the DBS patient population does not meet. Recording modality is therefore not a secondary detail but a hard constraint on translation, however well a biomarker performs on LFP, it can only benefit the subset of patients whose hardware and clinical circumstances allow that signal to be recorded at all. Scalp EEG offers substantially wider access, whole-brain coverage, and dramatically lower cost, at the price of spatial resolution and signal-to-noise ratio (SNR). Whether it can support DBS-biomarker work at all has been directly contested.

The historical objection rests on the artefact. Scalp electrodes record the subcortical stimulation pulse superimposed on the cortical potentials of interest, and standard artefact-subtraction methods rest on assumptions that do not consistently hold. Miocinovic et al. (2018) demonstrated that evoked-potential amplitudes vary by more than 25% between opposite-polarity stimulation settings in the majority of recordings, undermining the linear-subtraction approach on which many earlier studies relied (e.g. Sun and Hinrichs, 2016). Walker et al. (2012) had earlier restricted their EEG analyses to symptomatically ineffective low-frequency stimulation precisely because artefact contamination was tractable at those frequencies but not at clinically effective high-frequency settings, though they did go on to demonstrate short-latency cortical activation during clinically effective STN and thalamic DBS through careful paradigm design. Hale et al. (2026), reviewing the field's cortical evoked-potential work, note that EEG nonetheless remains the most widely used non-

invasive modality in this literature, employed in many cortical evoked-potential studies owing to its cost, accessibility, and non-invasiveness.

The contradiction between EEG being too artefact-prone for reliable use and EEG remaining the field's most widely used non-invasive modality is one of paradigm rather than of modality. The artefact objection applies most forcefully to pulse-locked evoked-potential methods, which must isolate a cortical response time-locked to each stimulation pulse. Approaches that compare ongoing cortical activity across stimulation configurations largely avoid it, since a residual pulse artefact of realistic magnitude cannot produce a sustained statistical difference between recording blocks lasting minutes. Calvo Peiro et al. (2025) demonstrates this directly, discriminating cortical responses to DBS contact changes with stimulation amplitude held constant, a condition under which an amplitude-scaling artefact cannot account for the result. Scalp EEG is therefore a defensible modality for questions about sustained differences between configurations, while LFP and electrocorticography (ECoG) remain the clearest source of evidence on millisecond-scale responses to individual pulses (Miocinovic et al., 2018; Walker et al., 2012). The question this dissertation asks of the modality is whether scalp EEG, despite its lower fidelity, carries enough information to identify a patient's ON/OFF medication state.

2.5 Machine learning and deep learning for biomarker discovery

This section reviews ML and deep learning approaches to biomarker discovery, before turning to explainable AI specifically, the technique on which this dissertation's contribution centres. The shift from hand-crafted spectral features to learned representations has changed the shape of the biomarker-discovery problem in DBS. Merk et al. (2022a) framed the field's current position in a review that argues the case explicitly, ML based decoding can extend adaptive DBS beyond the single-biomarker (predominantly beta) paradigm on which the earliest clinical successes were built, integrating multivariate feature sets that no manual analysis could feasibly weigh together. Their companion empirical work (Merk et al., 2022b) provides a concrete example, showing that gradient-boosted decision trees applied to intraoperative ECoG decoded grip force more accurately than the same models applied to subthalamic LFP in the same eleven PD patients, a finding directly relevant to the modality argument above, in that it reframes the LFP-versus-cortical question as an empirical one about decoding performance rather than a settled one in LFP's favour.

Two broad approaches divide the field. Classical ML applied to engineered features (spectral power in canonical bands, phase-amplitude coupling, burst statistics) has strong performance on well-defined per-patient classification tasks and the interpretive advantage that its features are physiologically meaningful in advance. Deep learning takes the opposite starting point, rather than committing to a fixed set of hand-designed features, it operates, principally via convolutional neural networks (CNNs), on minimally pre-processed signals, and can, in principle, discover features that human designers would not have thought to specify. The canonical architectures, the compact EEGNet of Lawhern et al. (2018) and the deep and shallow ConvNets of Schirrmeister et al. (2017), were designed specifically to make CNNs practical on the small, noisy datasets characteristic of EEG research, and both have been widely adopted in DBS-adjacent work. Calvo Peiro et al. (2025), using a compact CNN of this class on scalp EEG during DBS parameter changes, argue that minimally pre-processed

input is preferable in their setting precisely because the classifier is not committed in advance to a particular spectral band or feature.

An advantage in a research question where the identity of the discriminating feature is itself part of what is being investigated. Their result is also informative on a separate methodological question, discriminating stimulation states from EEG segments as short as one second, across thirty independently trained models, at an average accuracy of approximately 78%, indicates that clinically relevant information is present even in windows too brief for classical spectral-power estimation to resolve reliably. This is the premise on which the present project builds directly.

Two limitations attach to this approach. The first is sample size, classifiers in this literature are typically built on cohorts of ten to twenty patients, at which conventional inference does not license population-level claims. Roy et al. (2019), reviewing 156 deep-learning-on-EEG papers, found the problem to be field-wide, with small single-site cohorts and a majority of studies impossible to reproduce for want of available code and data. Independently trained per-patient or per-hemisphere models mitigate this by functioning as internal replications, since consistency across twenty independently trained models is a different evidentiary claim from a single model on a small sample, but they do not license generalisation to the population.

The second is interpretability. A classifier that discriminates two states has demonstrated only that information about the difference is present in the signal, which is a necessary but not sufficient step towards a biomarker. Explainability techniques, including feature ablation, saliency methods and physiologically grounded post-hoc analyses, are the mechanism by which detection is converted into a defined feature that can be inspected and validated on its own terms. Explainability in the DBS machine-learning literature remains shallow, dominated by feature ablation rather than mechanistic characterisation of what a network has learned, and this is the specific gap the present work addresses. A third constraint identified by Roy et al. (2019), the absence of shared benchmarks and standardised evaluation protocols, remains largely unaddressed and limits how comparable published results can be.

2.6 Translation and the outcome gap

A biomarker's sensitivity to a parameter change is not the same as that change being clinically beneficial, and conflating the two is the most consequential weakness in the literature surveyed above. Detection accuracy is an intermediate endpoint, a classifier that distinguishes contact A from contact B has shown that the cortex responds differently to the two, not that either is therapeutically superior, nor that a system driven by that classifier would converge on better settings than a clinician would.

Almost the entire alternative-biomarker literature sits at the detection or correlation stage. Mid-gamma and finely tuned gamma work, cortical beta-burst dynamics, and the EEG-based deep-learning classifiers just discussed are validated by their ability to detect a stimulation or medication state, or by their cross-sectional correlation with symptom severity at a single time point, not by prospective clinical outcome under the biomarker's own control. Calvo Peiro et al. (2025) are explicit about this in their own case, their work demonstrates that small parameter changes are detectable at the scalp, while defining the biomarker and correlating it

to patient symptoms is deferred to future work. That candour is appropriate, and it applies with equal force to the wider field.

A signal can be highly sensitive, reproducible, and mechanistically interesting while remaining clinically unproven. Sensitivity is often mistaken for readiness precisely because it is the property easiest to measure.

Only beta has crossed from detection into outcome-level trials, and its passage illustrates both the possibility and the difficulty of that transition. Oehrn et al. (2024) found, in a blinded, randomized crossover trial, that adaptive stimulation improved motor time and quality of life relative to optimised conventional stimulation. Tanimura et al. (2025), in a subsequent blinded crossover trial, found no statistically significant difference between the two, with directional effects split across endpoints, ON-time favoured conventional stimulation, while dyskinesia and UPDRS scores favoured adaptive. Busch et al. (2025), reporting real-world use of chronic adaptive stimulation in eight patients, found improved overall well-being at the group level but a significant within-subject benefit for general movement in only three of eight. That the biomarker with a decade of validation behind it still yields ambiguous outcome data should temper expectations for the newer candidates, none of which has been subjected to comparable scrutiny. Detection-stage work is a necessary precondition for everything that follows, the lesson is that the field's confidence in its candidate biomarkers currently rests on evidence that stops short of the clinical question those biomarkers are meant to answer. This dissertation sits at that same detection stage, it asks whether scalp EEG contains interpretable, physiologically meaningful information about medication state at all, leaving the correlation of any such marker with clinical outcomes to future work, consistent with the field's current state of evidence rather than in advance of it.

2.7 Conclusion and project positioning

Three limits emerge from the trajectory traced above. First, the incumbent biomarker is deployed but incomplete, beta is restricted to patients with sensing-capable devices, present in fewer configurations than trial numbers alone suggest, and drifts only slowly enough that a single-timepoint threshold cannot be assumed valid at the next. Second, the alternative candidates that have accumulated the strongest mechanistic evidence, gamma-band signatures and cortical evoked responses, remain almost universally validated against detection rather than outcome, and the machine-learning approaches that now increasingly mediate that detection compound the problem by adding a second layer of features that are difficult to inspect. Third, the modality with the widest patient access, scalp EEG, has been treated principally as a poor substitute for LFP rather than as a modality suited to a different class of biomarker question, one about sustained difference between stimulation configurations rather than millisecond-scale responses to individual pulses.

This dissertation is positioned at the intersection of the second and third gaps. It applies explainable AI (XAI) to a deep-learning model of scalp EEG trained to discriminate medication state, in order to characterise the cortical features driving that discrimination. The premise is not that beta is wrong, but that it is one hand-selected feature among many that a method unconstrained by prior commitment to a particular band could in principle surface.

The project sits at the transition from detection to characterisation and does not attempt to close the outcome gap identified in Section 2.6.

3. Methodology

3.1 Cohort, inclusion and exclusion, and session structure

The recordings analysed here are secondary data from the Living Lab study (Crook-Rumsey *et al.*, 2023), a cross-sectional observational study of cognition and behaviour in people living with neurodegenerative and other neurological conditions, conducted at the UK Dementia Research Institute Care Research and Technology Centre, Imperial College London. Participants attended for a single visit and carried out activities of daily living within the Living Lab, a 32 m² apartment-laboratory furnished with a working kitchen, dining area, sitting room, bed and bathroom, which provides an intermediary setting between a conventional laboratory and the home. Scalp electroencephalography was recorded throughout using a 21-channel DSI-24 dry-electrode system (Wearable Sensing, California, USA), which transmits wirelessly and permits free movement during the functional tasks. The parent study was approved by the Imperial College Research Ethics Committee (reference 21IC6992) and the Imperial College Research Governance and Integrity Team and was conducted in accordance with the Declaration of Helsinki. All participants gave informed consent. No new data were collected for this project, and no participant was contacted at any stage. The present work analyses only the Parkinson's disease cohort.

Each participant attended a single visit and completed the task battery twice, once after levodopa had taken effect, and once after it had worn off. The two blocks are stored as separate files, A for the medicated (ON) state and C for the wearing-off (OFF) state.

Pharmacology fixes the order of the blocks, since the wearing-off state can only be reached by waiting after the medicated one. ON therefore always precedes OFF and the order cannot be counterbalanced. Medication state is consequently confounded with elapsed time within the visit, with contact drift and with fatigue, in the same direction for every participant. The validation design in Section 3.6 was built around this constraint from the start, before any result had been seen.

Labels were assigned at the level of the recording, and every window inherits the label of its block. No onset buffer was applied. A buffer is appropriate when a recording starts at the moment of dosing and crosses into the medicated state part-way through. Here recording began only after the waiting period had elapsed. The risk that a slow responder had not reached full effect is carried as label noise. The label indexes the intended state, not a measured one. Restricting the analysis to participants with verified dose timing or a concurrent UPDRS assessment was considered and rejected, assessments in both states exist for only 11 of the 35 participants.

The analysed cohort comprises 35 participants, each contributing one session per state: 70 sessions and 37,174 windows, of which 20,122 come from block A and 17,052 from block C. The cohort was fixed before any modelling experiment and was never revised. An invariant check aborts any run that fails to reproduce those totals.

Reduction to this cohort proceeded in four stages, shown in Figure 3-1. Recordings made under a different protocol with the same DSI-24 headset were excluded on grounds of provenance. Participants without a recording in both states were excluded, two were never processed, one carrying an incompatible 13-channel montage and one an unresolvable file ambiguity and six failed the quality-control criteria fixed in the audit configuration and applied uniformly. Three participants omitted from early versions of the cohort by the file-selection defect of Section 3.2 were subsequently recovered, so the final count reflects a corrected fault rather than a post-hoc quality judgement.

Two analysis sets are used and are never merged, namely, a task-split subset and a wider cohort. The task-split subset carries the main analysis of Chapter 4. The wider cohort is the 35 participants described above, and it carries the exploratory leave-one-patient-out work of Section 4.6. The task-split subset is the 27 participants for whom both kitchen tasks, the kettle task and the toast task, could be recovered in both sessions with disjoint spans and at least five evaluation windows each. The remaining eight participants were excluded by this criterion alone, recovery failures were distributed across both sessions and not concentrated in one, so the reduction does not select for a particular medication state.

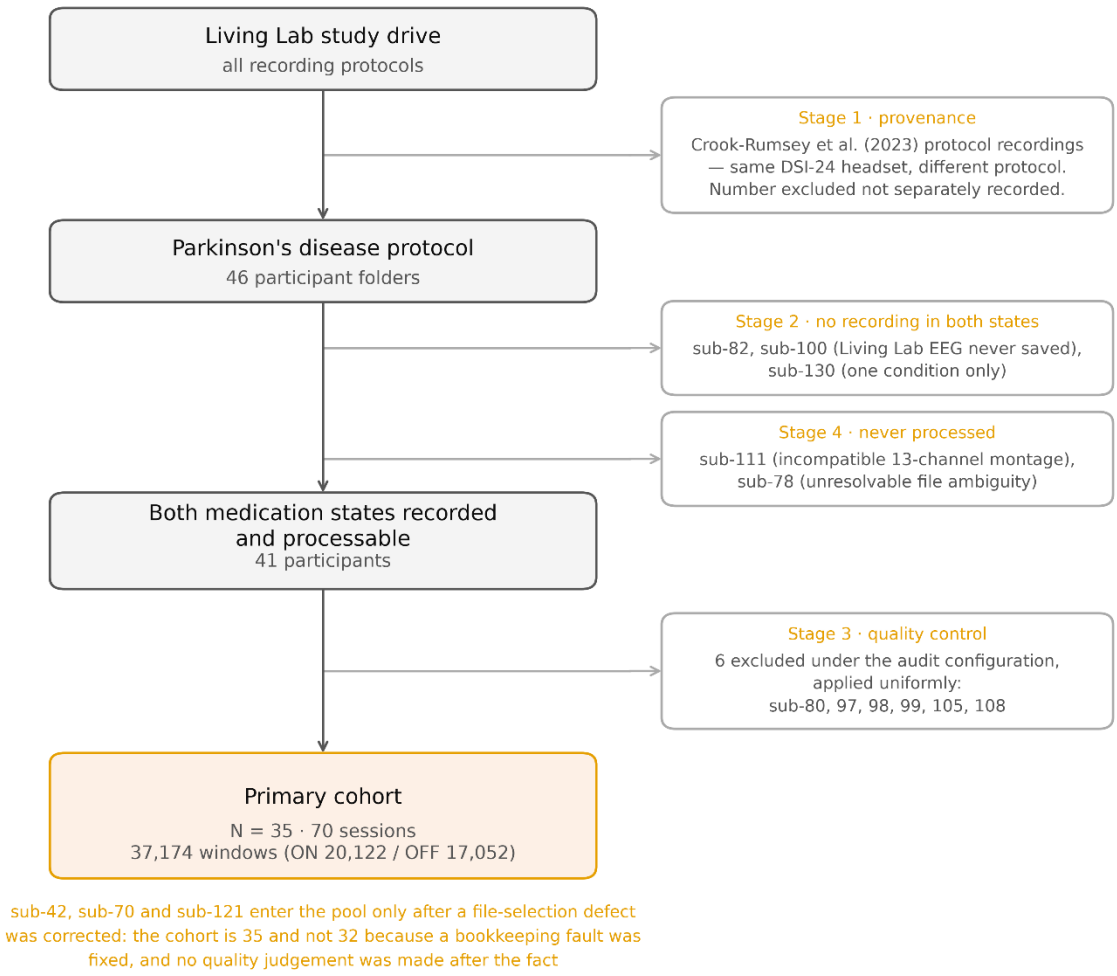


Figure 3-1. Participant flow from the study drive to the primary cohort. The four reduction stages described in Section 3.1 are shown as separate exclusions. Three participants had no recording in both medication states (stage 2); two were never processed, one for an incompatible 13-channel montage and one for an unresolvable file ambiguity (stage 4); and six failed the quality-control criteria fixed in the audit configuration and applied uniformly (stage 3). The number excluded on provenance is not separately recorded in the study documentation. Counts reconcile exactly at each step (46 - 3 - 2 = 41; 41

- 6 = 35). The cohort is 35 rather than 32 because a file-selection defect that had omitted three participants was corrected, and no quality judgement was made after the fact. Section 4.1 and Figure 4-1 carry the further reduction to the 27-participant task-split subset.

3.2 Acquisition and ingestion

Recordings were acquired at 300 Hz. Dry electrodes make ambulatory recording practical, but they bring higher and less stable impedance and a heavier artefact burden than wet electrodes (Hinrichs *et al.*, 2020). The amplifier records referentially against Pz, which leaves that channel degenerate. A1 and A2 are ordinary channels here, not a hardware reference, and re-referencing consumes them.

The accelerometer triplet is present but unusable, all three channels sit at approximately 1 g (gravitational acceleration) throughout every recording, indicating no detectable movement was captured at any point. Without it, movement can only be estimated from the EEG itself, and any such estimate is contaminated by the activity it is meant to quantify.

Sessions were ingested from the raw comma-separated value (CSV) exports rather than the European Data Format (EDF) files, since EDF exists for only 39 of the 89 sessions and two of those are unreadable. The two formats were verified equivalent before the decision was taken.

3.3 Preprocessing pipeline

Preprocessing proceeds through eight stages in a fixed order that cannot be altered at run time, re-referencing, band-pass filtering, notch filtering, independent component analysis (ICA), channel selection, resampling, windowing and normalisation. Table 3-1 gives the parameters, all of which are read from a single validated configuration file rather than appearing as literals in the processing code.

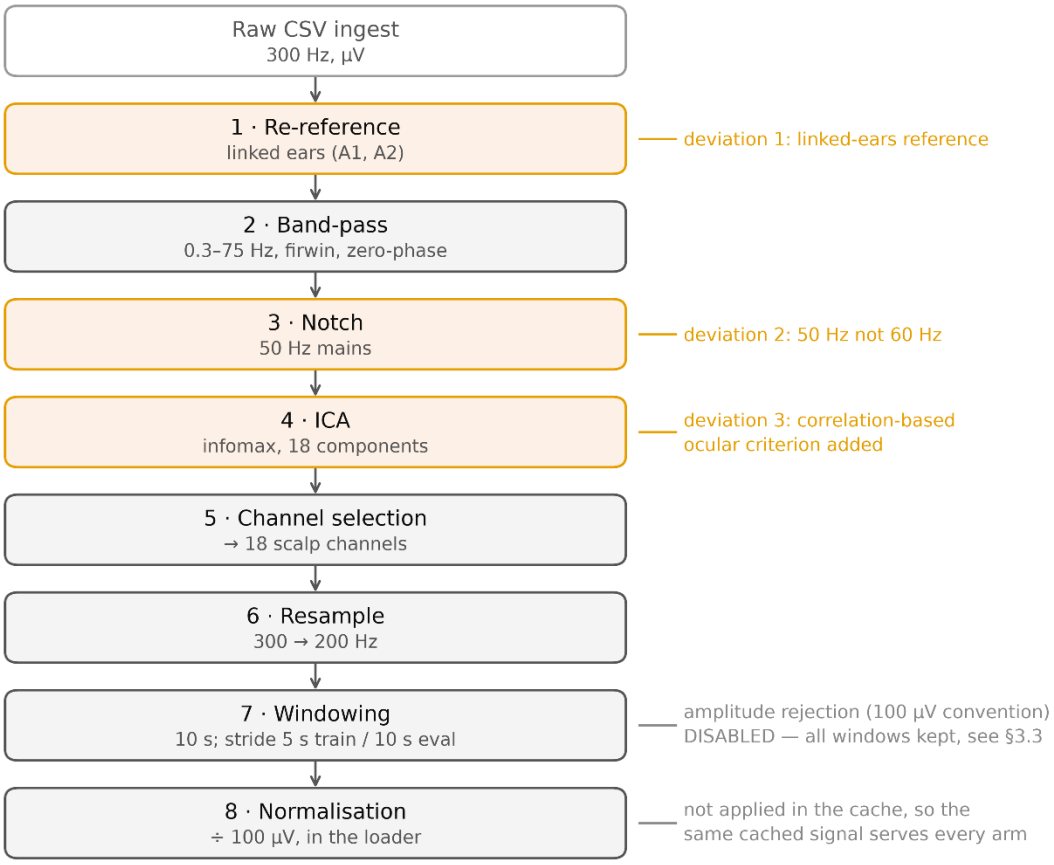


Figure 3-2. The preprocessing pipeline. The eight stages are numbered in the order fixed in code, which cannot be altered at run time. Parameters are those of Table 3-1 and are read directly from the configuration file that produced the cached signal. Stages shown in orange carry the three deviations from the published CBraMod convention justified in Section 3.3, the linked-ears re-reference, the 50 Hz rather than 60 Hz notch, and the correlation-based ocular criterion added to ICA. Raw CSV ingest is the input to the pipeline rather than one of the eight stages and is shown unnumbered. Amplitude rejection is drawn to the side rather than in the chain because the convention's 100 uV threshold is disabled here and no window is discarded; Section 3.3 and Figure 3-4 give the reason. Normalisation is applied in the loader rather than written into the cache, so every experimental arm consumes byte-identical cached input.

Table 3-1. Preprocessing parameters.

Stage	Setting
Reference	mean of A1 and A2 (linked ears), applied first, before channel selection and filtering (deviation from CBraMod, which does not re-reference)
Band-pass	0.3–75 Hz, finite impulse response, firwin design, zero phase
Notch	50 Hz (UK mains, pretraining convention specifies 60 Hz) (deviation from CBraMod))
ICA	extended infomax, 18 components, fixed seed, max 500 iterations
ICA fit copy	1 Hz high-pass, resampled to 100 Hz; unmixing applied to full-rate data

Stage	Setting
ICA removal, criterion 1	winning ICLabel class in {muscle artifact, heartbeat, line noise, channel noise}, probability ≥ 0.5
ICA removal, criterion 2	correlation ≥ 0.5 with Fp1 or Fp2 within 1–10 Hz, treated as ocular (added beyond CBraMod's ICLabel-only convention; see below)
Channels retained	Fp1, Fp2, F7, F3, Fz, F4, F8, T3, C3, Cz, C4, T4, T5, P3, P4, T6, O1, O2
Channels excluded	Pz (degenerate original reference); A1, A2 (consumed by re-referencing)
Resampling	300 Hz to 200 Hz, after ICA and channel selection
Window	10 s, 2000 samples per channel, layout (batch, 18, 10 patches, 200 samples)
Stride	5 s training (50% overlap); 10 s evaluation (no overlap)
Normalisation	division by 100 μV, applied in the loader rather than the cache

Table 3-1. Preprocessing parameters. Parameter values for each of the eight pipeline stages shown in Figure 3-2, read directly from the validated configuration file that produced the cached signal; all values are those applied, not illustrative defaults.

CodeBrain (Ma et al., 2026) inherits its preprocessing convention from Criss Cross Brain Model (CBraMod) (Wang et al., 2024) (, and Table 3-1 reproduces the published CBraMod pipeline except where noted. Matching that convention matters more here than it would for a model trained from scratch, a pretrained backbone learned its representation from inputs with particular statistics, so any departure from those statistics is a distribution shift. The convention was taken from the publication and not from the released code, which applies no filtering or resampling and works on bipolar derivations. Three deviations were made, and each is justified below.

Re-referencing

Every channel carries a common Pz term, so subtracting the mean of two channels that also carry it cancels that term algebraically. Re-referencing is applied first because A1 and A2 are dropped at channel selection, and because working on unfiltered data keeps the reference independent of the filter's edge behaviour. The acquisition-level mismatch cannot be repaired retrospectively, the pretraining corpus was recorded against standard clinical references and never against Pz, so no re-referencing scheme reproduces the spatial covariance the model was trained on. Mean-of-A1/A2 (linked ears) was adopted as the closest available approximation, being a standard symmetric reference free of single-site bias and computable before A1 and A2 are consumed. Chapter 5 treats this mismatch as one candidate contributor to the domain gap rather than as a claim tested here.

Filtering

The band-pass (0.3–75 Hz) follows the published CBraMod convention, whereas the notch frequency was changed. The pretraining corpus was recorded where mains frequency is 60 Hz, but these recordings were made in the United Kingdom at 50 Hz. The notch was therefore set to 50 Hz, applying the published 60 Hz value would have left the real 50 Hz interference in the signal while filtering a band with no interference to remove.

ICA and the failure of automatic component labelling

The number of components equals the number of scalp channels, so the decomposition is full-rank and removing zero components is exactly the identity. Any before-and-after comparison can therefore be attributed to the removal itself and not to rank reduction.

The second removal criterion exists because the first fails on this data. ICLabel, an automatic classifier that assigns each independent component to one of seven source classes, identified only one or two ocular components per 50-minute recording when used alone. Blinks were counted manually at 20 to 27 per minute in sub-131 and 4 to 13 per minute in sub-134, the lower figure consistent with the reduced spontaneous blink rate recognised in Parkinson's disease. ICA had separated the blinks correctly, producing components whose time courses tracked the frontal channels almost exactly, but ICLabel assigned these eye-class probabilities of 10% to 25% and a winning label of brain or other. The pipeline acts on the winning label, so no probability threshold would have recovered them. ICLabel was trained on seated, common-average-referenced recordings, whereas these are ambulatory and referenced to linked ears, either difference could shift the features on which its classification depends.

The correlation criterion recovers exactly these components. It flags independent components 16 and 17 in both sessions of sub-134 and agrees with ICLabel in sub-131. One limitation is carried forward unresolved. ICLabel contributed no non-ocular removals in any recording that was inspected, and the available evidence cannot establish whether that is because no such components were genuinely present, or because ICLabel failed to flag components that were. Appendix A records two detector behaviours that make the 0.5 threshold unsafe to transfer.

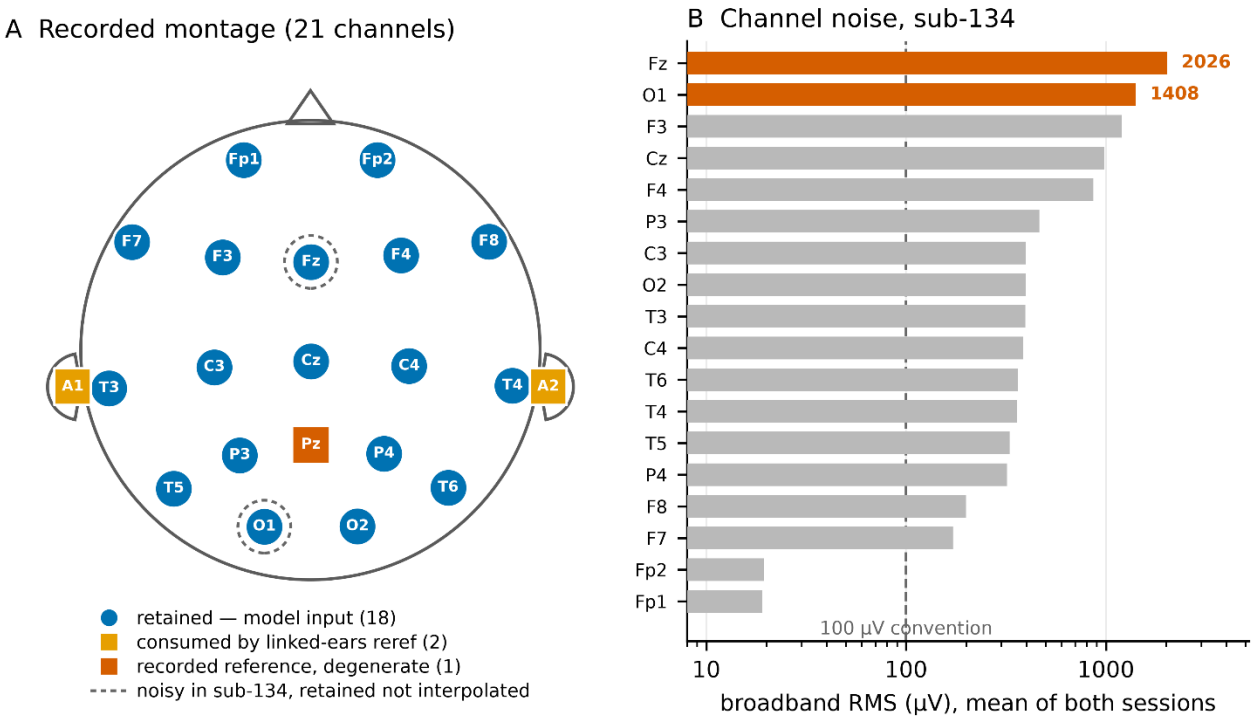


Figure 3-3. The recorded montage and the decision to retain rather than interpolate noisy channels. (A) The 21 recorded DSI-24 positions on a 10-20 schematic, divided three ways, the 18 scalp channels the model consumes, A1 and A2, which are ordinary channels here and are consumed by the linked-ears re-reference, and Pz, the recorded reference, which is degenerate and carries no signal. Positions are the true standard_1020 coordinates under an azimuthal-equidistant projection. Fz and O1 are ringed as the two channels identified in Section 3.3 as chronically noisy in sub-134. (B) Broadband RMS for each of the 18 retained channels in sub-134, the mean of that participant's two sessions, on a logarithmic axis with the convention's 100 uV threshold marked. Fz and O1 exceed every other channel by a wide margin, in 2026 and 1408 uV. Both are nonetheless retained rather than interpolated, because the model's positional encoding operates over a fixed channel grid and the montage must therefore be identical across the cohort. Panel B covers sub-134 alone; the per-channel amplitude survey was run on two participants only, so no cohort-wide version of this panel is available.

Channel selection and amplitude rejection

The montage is fixed across the cohort, since the positional encoding operates over a fixed channel grid. Sub-134's chronically noisy O1 and Fz were therefore retained rather than interpolated. The convention's 100 µV rejection threshold discards 98% of windows across the cohort, and between 85% and 100% of them in any individual recording. That figure was checked and is genuine, not the result of a unit error. Rejecting at patch level instead is more permissive, but not nearly enough to rescue the threshold, survival at the convention rises only from 1.6% to 8.4%. The threshold sweep was run to characterise the regime, not to find a workable replacement, and it showed that no replacement exists, any threshold loose enough to retain usable windows also retains the artefacts the threshold is there to remove. All windows are therefore kept, and no claim made later depends on the signal being artefact-free.

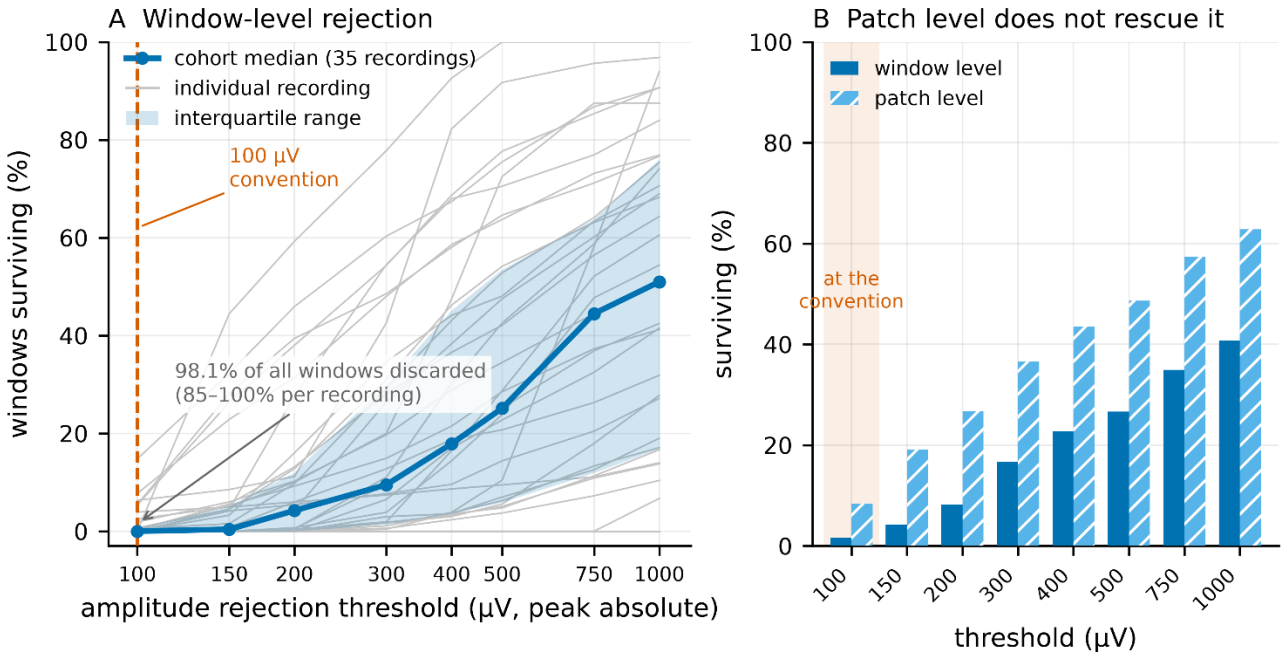


Figure 3-4. Window survival against the amplitude-rejection threshold. (A) The fraction of windows surviving rejection at each threshold, one thin grey line per recording, with the cohort median and interquartile range overlaid and the convention's 100 uV value marked. At that threshold 98% of all swept windows are discarded, and no recording retains more than 15% of its own. (B) The same sweep applied at patch level rather than window level, which is the obvious remedy. Patch-level rejection is consistently more permissive, but it does not rescue the threshold, survival at the convention rises only from 1.6% to 8.4%. The curves have no knee, so there is no value at which usable windows are retained and artefacts are not; the sweep therefore characterises the regime rather than locating a workable replacement. All windows are consequently kept, and no claim in this dissertation depends on the signal being artefact-free. The sweep samples 250 to 300 windows from each recording, 15,752 in total, rather than covering the full 37,174-window cohort.

Windowing and normalisation

Ten seconds is the longest window that fits inside the shortest task, which lasts about 12 s. Overlapping windows in training act as augmentation, since EEG is quasi-stationary over these timescales. Using non-overlapping windows at evaluation is a requirement and not a preference, treating dependent windows as independent narrows confidence intervals that ought to be wide. Per-session normalisation was rejected because a session is exactly one medication state, any statistic computed over a session and divided out would remove the between-state difference along with the nuisance variance it was intended to suppress. Standardisation of any kind similarly destroys absolute amplitude structure, which this study cannot afford to discard given that amplitude is itself a candidate marker of medication state. Division by a fixed constant (100 µV) was therefore adopted in place of per-session or per-window standardisation, and the two standardisation schemes are retained instead as separate arms in Table 3-2.

Task-label recovery

Task boundaries appear only in separate annotation files. None of the 89 audited sessions carries usable trigger information, so boundaries can be established only by reconciling two clocks, which is a fitting procedure with its own failure rate. Window extraction therefore tiles each session end to end, independent of task boundaries, so cohort membership never depends on clock reconciliation succeeding. The secondary, task-split design carries no such constraint, task labels are recovered after the fact into a read-only sidecar file and attached to the existing windows, leaving the cached windows and manifest untouched.

A labelling failure here narrows the task-split subset rather than affecting the wider cohort. The search evaluates candidate offsets across a range of ±4 hours, selects the offset that recovers the most tasks, and breaks ties toward the candidate closest to a whole hour.

3.4 Model architecture and adaptation strategy

CodeBrain (Ma et al., 2026) is an EEG foundation model, pretrained by masked reconstruction over a large corpus of clinical recordings. Only the EEGSSM backbone is used downstream. It has eight residual blocks, each combining a structured global convolution with single-token sliding-window multi-head attention, and approximately 15.2 million parameters in total. The vector-quantised codebooks are artefacts of the tokeniser and are not loaded at all This was verified by confirming that the codebook projection weights remained byte-identical to the pretrained checkpoint after full fine-tuning. The classification head was configured for 18 channels, where the released configuration hardcodes 16. The backbone accepts this because its kernels are truncated to whatever sequence length is presented. An alternative flattened head carrying 72 million parameters overfitted on this data and was replaced with a pooled one.

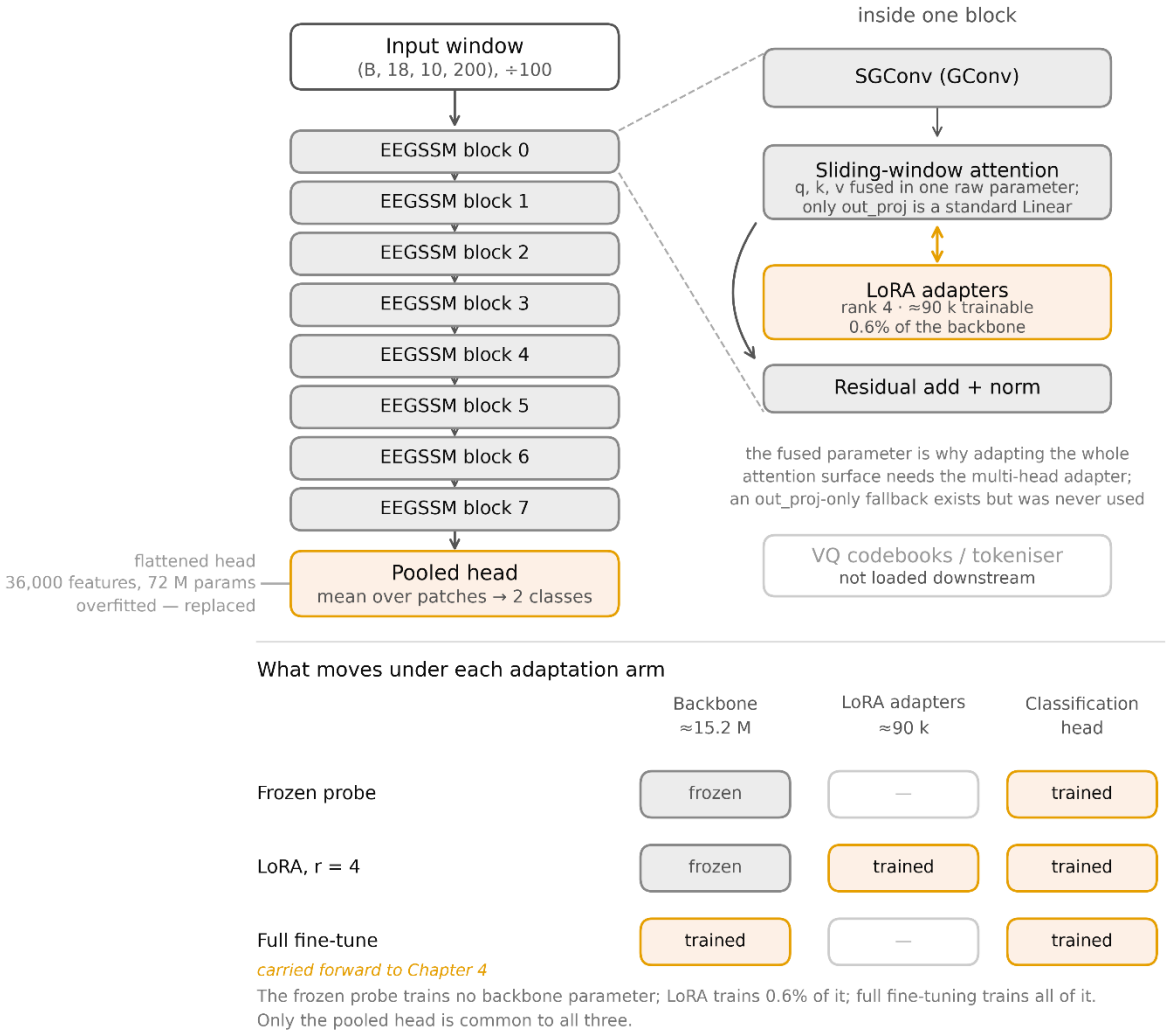


Figure 3-5. The CodeBrain backbone and the three adaptation arms. Left, the data path. A 10 s window enters as a (batch, 18 channels, 10 patches, 200 samples) tensor divided by 100 uV, passes through the eight EEGSSM residual blocks, and is

read out by a pooled head that averages over patches. The flattened alternative shown in grey, which would have carried 36,000 features and 72 million parameters, overfitted on this data and was replaced. Right, the interior of one block. Each combines a structured global convolution with single-token sliding-window multi-head attention. The query, key and value projections are fused into a single raw parameter and only the output projection is a standard linear layer, which is why adapting the whole attention surface requires the dedicated multi-head adapter rather than the output-projection fallback, and why LoRA is injected into the attention module rather than appearing as a step in the data path. The vector-quantised codebooks belong to the tokeniser and are not loaded downstream at all. Bottom, which parameters move under each of the three adaptation arms. The frozen probe trains no backbone parameter, LoRA trains roughly 90,000 of them at rank 4, and full fine-tuning trains all 15.2 million. Only the pooled head is common to all three. Full fine-tuning is the arm carried forward to the main analysis and the attribution work of Chapter 4.

Low-rank adaptation (LoRA) inserts trainable low-rank matrices alongside frozen weight matrices, so a pretrained model can be adapted by training a small fraction of its parameters. LoRA was designated the primary adaptation method on a criterion fixed before any fine-tuning result had been seen. With 35 participants, the binding constraint is the ratio of trainable parameters to independent units of evidence, which makes a method that adapts under one percent of the backbone, and leaves the pretrained temporal filters intact, the appropriate default. Full fine-tuning was kept as a comparison arm rather than promoted to the primary method. Its purpose is to bound what extra capacity can buy, and so to rule out insufficient adaptation as an explanation for a weak result.

Insertion points were established by inspecting the live module tree, not assumed from the architecture description. The eight blocks fuse the query, key and value projections into a single raw parameter, and only the output projection is a standard linear layer, so adapting the full attention surface requires the dedicated multi-head adapter. A fallback targeting the output projections alone was implemented but never used. The convolution kernels and the tokeniser path stay frozen throughout. Rank was searched rather than fixed by hand. The search returned rank 4, a learning rate of 5×10^{-4} and dropout 0.1, giving roughly 90,000 trainable parameters, or 0.6% of the backbone. That is the low-capacity corner of the search space.

3.5 Experimental configurations and hyperparameter search

Hyperparameters were selected by random search over ten configurations, each evaluated by 5-fold cross-validation on the wider cohort, grouped on participant so that no participant appears on both sides of an inner split. This grouping rule applies to the wider**,** cross**-**patient design only, the task-split design splits within participant by construction and is evaluated separately. Table 3-2 gives the search space and every configuration trialled.

Table 3-2. Configurations trialled.

Family	Configuration	Setting	Role
Search space	rank	{4, 8, 16}	10 random configurations

Family	Configuration	Setting	Role
Search space	learning rate	$\{5 \times 10^{-4}, 1 \times 10^{-3}, 2 \times 10^{-3}\}$	
Search space	dropout	{0.0, 0.1, 0.25}	
Selected	LoRA	r = 4, lr 5×10^{-4}, dropout 0.1	pre-registered arm (Section 4.6)
Arm	frozen probe, mean-pooled	200 features, no backbone training	representation floor
Arm	frozen probe, flattened	36,000 features	pooling control
Arm	full fine-tuning	all ≈15.2 M parameters, lr 1×10^{-4}	capacity bound
Arm	LoRA, per-window global standardisation	r = 16, lr 2×10^{-3}, dropout 0.0	input-representation arm
Arm	LoRA, per-channel standardisation	as selected	input-representation arm
Arm	hybrid, backbone plus classical features	concatenated	test for complementary information
Control	classical relative band power	Welch, 2 s segments, 0.5 Hz resolution; δ 0.5–4, θ 4–8, α 8–13, β 13–30, γ 30–45 Hz	accessibility to conventional analysis
Control	recording-duration heuristic	window count only, no EEG	trivial floor
Training	optimiser, schedule	AdamW, weight decay 0.01, cosine with 10% warmup, batch 64, ≤20 epochs, bf16	all trained arms
Training	early stopping	inner-validation balanced accuracy, patience 4	all trained arms

Table 3-2. Configurations trialled. The hyperparameter search space and the ten random-search configurations evaluated by grouped 5-fold cross-validation, together with the selected configuration and every experimental arm and control derived from it. The Role column states the purpose each arm or control serves in the capacity ladder described in Section 3.5.

The arms in Table 3-2 form a capacity ladder. The frozen probe establishes the representation floor, low-rank adaptation and full fine-tuning increase the trainable parameter budget in turn, and the standardisation arms and the classical control test whether the input representation rather than the adaptation capacity is the limiting factor. All three adaptation arms were

evaluated under both validation designs of Section 3.6. The arm carried forward to the attribution analysis of Section 3.7 was selected on main-analysis performance, reported in Section 4.2.

The search has two limitations, both declared here. First, inner-validation scores across the ten configurations span a range comparable to the noise at this sample size, so selection is partly selection on noise. The standardisation arm settled on the highest-capacity, zero-regularisation corner of the space where the raw-input arm settled on the lowest therefore differs from its comparator in both input representation and hyperparameters and does not cleanly isolate the effect of normalisation.

Second, the search was run once over the pooled participants with grouping and not repeated inside every outer fold. Literal nesting was estimated at roughly 330 hours of compute against 15 for the procedure used. Selection therefore saw every participant's data, which makes the cross-patient estimates for the adapted arms optimistic by an amount that is not quantified. The classical control was run under true nesting, its nested and non-nested balanced accuracy differed by 2.8 percentage points (0.629 nested against 0.657 non-nested), which is the only direct evidence available on how large that optimism is likely to be for the adapted arms.

3.6 Validation design and evaluation metrics

Three validation designs were run over the course of the project. They are described here in the order in which the argument of Chapter 4 uses them, which is not the order in which they were run. The main design is a pooled task split on the 27-participant subset. One shared model is trained on windows drawn from one kitchen task, taken from both medication blocks, and tested on non-overlapping windows from the other kitchen task. Both directions are run and reported separately rather than averaged, since Chapter 4 shows they differ. The participant is the unit of analysis. Splitting by task rather than by time keeps elapsed time out of the split, so the fixed block ordering described in Section 3.1 cannot supply the answer through contact drift, fatigue or ambient interference accumulating over a session. Early stopping uses a trailing contiguous block of the training task, because overlapping windows would otherwise place near-duplicates on both sides of the split. One precondition was checked before the design was run, both kitchen tasks are logged in both blocks for 34 of the 35 participants, so task type is not confounded with medication state.

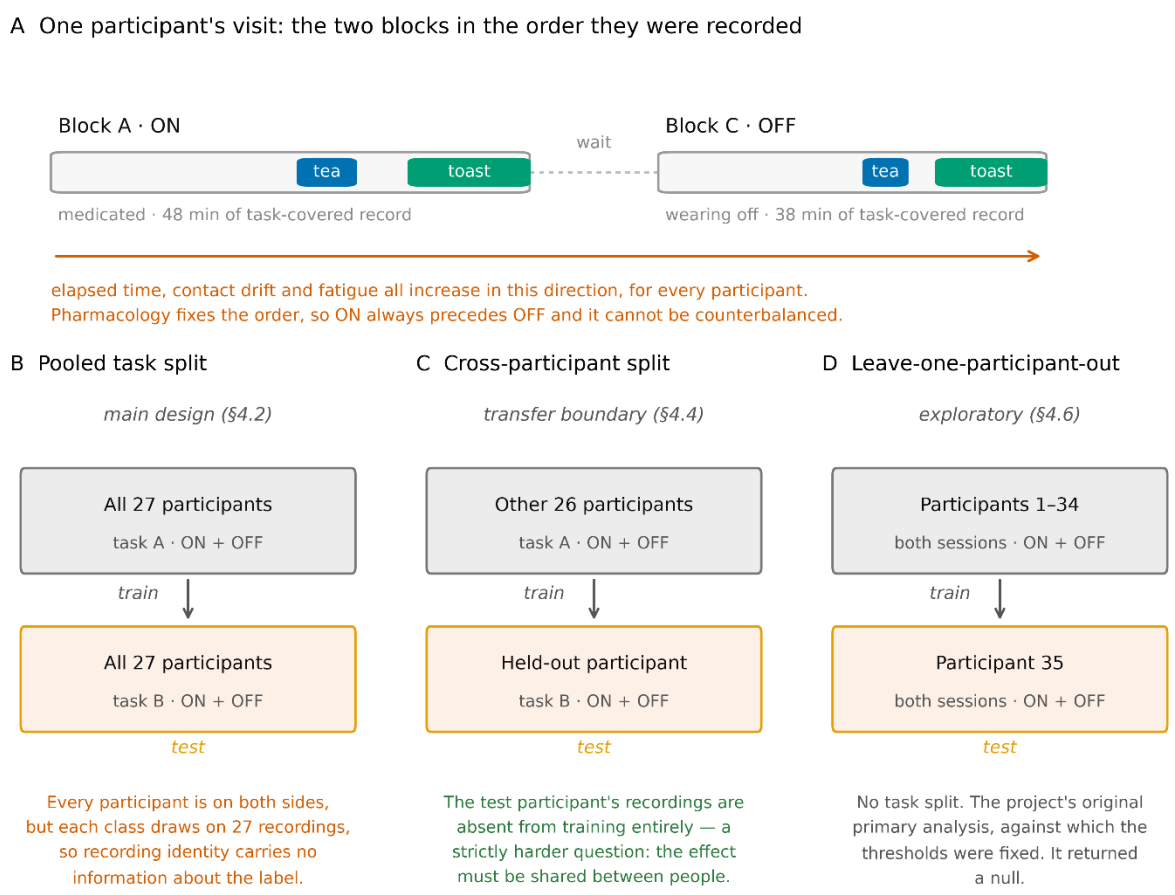


Figure 3-6. The session timeline and the three validation designs. (A) One participant's visit. The medicated block is always recorded first and the wearing-off block second, because the wearing-off state can only be reached by waiting; the order is fixed by pharmacology and cannot be counterbalanced. The two kitchen tasks are shown at their median positions within each block, recovered from the task-window sidecar across 62 sessions; tea precedes toast in 61 of them. Both blocks are drawn to the same scale, so the shorter OFF block reflects a real difference in recorded length. The interval between blocks is drawn as a break because it is not recorded in the window data. Everything that drifts over a visit, elapsed time, electrode contact and fatigue, therefore moves in the same direction as medication state, for every participant. (B) The main design, one shared model trained on all 27 participants' windows of one kitchen task and tested on their windows of the other. Every participant appears on both sides, but each medication class is assembled from 27 recordings rather than one fixed pair, so recognising an individual recording carries no information about the label. (C) The same protocol re-run with the test participant's recordings removed from training altogether, which asks the strictly harder question of whether the effect is shared between people. (D) The session-level leave-one-participant-out design used in the exploratory work of Section 4.6, in which both of a participant's sessions are held out and no task split is applied. This was the project's original primary analysis and the design against which the pre-registered thresholds were fixed.

Pooling is what makes this design interpretable. Each participant contributes exactly one recording per medication state, so within a single participant the medication label and the recording identity are the same variable, and any property that is stable within a recording and differs between recordings will answer the question. An earlier per-participant variant of the task split failed on exactly this route, as Section 4.2 reports. Under pooling, each medication class is assembled from 27 participants' recordings rather than from one fixed pair, so recognising an individual recording carries no information about the label, while generalisation across behavioural context is still required.

Duration matching is applied to the main design. Windows are equalised at K = min (ON, OFF) within each participant-by-task cell, which equalises the confounded ON/OFF pair and nothing else. Participants still differ from one another in window count, and the two kitchen tasks still

differ from each other. Matching at participant level or at task level was rejected, a single shared K collapses the sample, and per-task matching cuts the training set for the principal direction by 60%, which would confound removal of the duration effect with simple starvation of data. Matching at the chosen level costs about 10% of windows. Surviving windows are spread uniformly across each session's task span rather than taken contiguously, because under K = min (ON, OFF) a contiguous rule leaves the shorter session holding both task onset and task offset while the longer session keeps only its centre, which turns the presence of a task transition into a proxy for being the shorter session. Both matched and unmatched estimates are reported so that the cost of the correction is visible.

The second design tests transfer between individuals. It re-runs the pooled task split with the test participant's recordings removed from training altogether, other participants' windows of one task for training, the held-out participant's windows of the other task for evaluation. This is a strictly harder question than the main design asks, since it requires the medication-state effect to be shared across people rather than merely stable within a person across a change of task. Chapter 4 reports it as a boundary on the main result rather than as a test the main result must pass, and Chapter 5 discusses why a genuine but individually expressed drug response would fail it.

The third design is the session-level leave-one-participant-out cross-validation used in the exploratory work of Section 4.6. In each outer fold, both sessions belonging to one participant are held out entirely and the model is trained on the remaining 34. Predictions are aggregated to session level, the mean predicted probability of OFF across a session's non-overlapping windows is compared with a threshold of 0.5, giving 70 predictions, balanced by construction. Evaluation is at session level because windows within a session are not independent of one another. This design was the project's original primary analysis and the pre-registered thresholds of Section 3.8 were fixed against it. It returned a null, and Section 4.6 reports it in that role.

3.7 Interpretability and attribution procedure

Attribution was applied to the arm that scores highest in the main analysis, the pooled task-split arm, duration-matched, evaluated under full fine-tuning in the toast-to-tea direction. Its purpose is to characterise the frequency content the model's decisions depend on. Attribution was withheld from the exploratory leave-one-participant-out arms of Section 4.6, on the grounds that attribution methods answer the question of what drove a model's decisions and applied to a model performing at chance they return the structure of the noise, presenting the resulting maps as candidate neural markers would be a misuse of this dataset.

Training is non-deterministic, and a single run gives no way to distinguish a genuine attribution pattern from one specific to that seed's initialisation. The arm was therefore retrained under five seeds and evaluated for stability against a gate fixed in advance, set to flag at a spread of ±0.03 and to stop at ±0.05. Checkpoints record the git revision, the backbone checksum and the seed. Every figure derived from this arm carries a stability caveat.

Perturbation was preferred to gradient attribution because it answers the same questions causally. Each perturbation degrades one property of the input, and the resulting change in balanced accuracy is read as that property's contribution. Phase randomisation, which preserves the amplitude spectrum while destroying phase information, was included as a specific architectural test. The patch embedding computes the magnitude of a real fast Fourier transform of the input, discarding phase entirely. If so, one of the model's input pathways is phase-blind by construction, and this perturbation tests whether that pathway is load-bearing for the model's decisions. A head-weight readout and a codebook analysis were both trialled and then abandoned on measurement grounds.

All channel-resolved analysis was withdrawn. The electrode cap moves substantially during a recording on this setup, so an electrode label does not index a stable scalp location, and no channel-resolved statistic can separate a participant-specific physiological explanation from participant-specific electrode positions. One lesson from this is retained for any future occlusion analysis. A random-occlusion baseline is essential. Without one, the apparent transfer is almost entirely the generic cost of deleting channels and reading it as transfer inverts the conclusion.

Spectral dependence was characterised with a band-stop sweep. An initial version used 2 Hz notches and found no dependence on any band. That null was withdrawn, a 2 Hz notch removes only about 4% of the power between 1 and 45 Hz, so the result was a statement about notch width and not about the model. The sweep was rebuilt using total notch widths from 5 Hz to 18 Hz (not ± values), centred at positions from 1 Hz to 45 Hz, giving 469 bands, each scored against all five seeds. Every filter was validated for stopband attenuation and passband preservation before the sweep was run.

Width is confounded with everything else in the sweep, so two controls are applied within width, a within-width standardisation, and an isotonic regression, a curve fit constrained to be monotonic, relating accuracy change to the fraction of power removed. The isotonic fit is computed per width. A pooled fit was tried first and was found to be incorrect, at any given power value, it averages narrow bands against wide ones, leaving a residual that only restates the fact that wider notches hurt more. At widths of 8 Hz and above the best band sits on the sweep's lower boundary, so those edges are artefacts of where the search stopped and only the narrower widths can be interpreted. The whole sweep was repeated in the opposite train-test direction, on the reasoning that a shared drug effect should show up whichever task carries the test label.

No attribution output is certified as a biomarker unless it passes the gate fixed in advance and stated in Table 3-4. Certification requires a candidate to be rebuilt as a minimal explicit feature and tested on its own under the cross-participant task-split design of Section 3.6, in which the test participant's recordings are absent from training at every stage. A design of that kind is described here as leak-free, meaning that no information about the evaluation participant reaches the model during fitting, whether through shared windows, shared recordings or a selection step. The certification gate is deliberately more demanding than the main analysis, because a feature offered as a biomarker makes a claim about people in general rather than about one person measured twice. Chapter 4 reports the outcome of that gate.

3.8 Control checks, pre-registered thresholds and statistical testing

Every control was constructed against a specific alternative explanation. Table 3-3 pairs each one with the explanation it eliminates.

Table 3-3. Control checks, what each one is, and the explanation it rules out.

Control	What it does	If the result had come out the other way, the explanation it rules out
Channel-amplitude probe	Classifies medication state from 18 numbers, the mean absolute amplitude of each channel. No frequency content, no waveform, no spatial pattern.	A high score means an electrode-contact fingerprint is sufficient, so a trained model scoring high tells us nothing about the brain.
Negative control (shuffled labels, refitted)	Runs the entire pipeline 100 times with the ON/OFF labels shuffled inside each participant, refitting from scratch every time.	A score above chance means the pipeline manufactures signal from noise, and every other number in the study is uninterpretable.
Positive control (participant identity)	Asks the same representation to say which of the 35 participants a recording came from, training on ON windows and testing on OFF windows.	A score at chance means the representation carries nothing usable from these recordings, so a null on medication state proves nothing.
Raw band-power ceiling for the identity task	Measures how well the raw input supports the identity task, before any model is involved.	Without it the positive control has no scale, a score of 0.40 is impressive or unimpressive depending on what the input can supply.
Recording-duration heuristic	Classifies medication state from one number, the count of windows in the session. Uses no EEG at all.	A high score means session length alone answers the question, and any model scoring at or below it has demonstrated nothing.
Duration matching at participant × task level	Re-runs the analysis with ON and OFF window counts equalised inside each participant-by-task cell.	A large change under matching means the unmatched result was driven by the length difference rather than by the signal.
Channel-order comparison	Repeats the analysis under three electrode-to-input mappings, the study montage order, CodeBrain's reported order, and a random permutation.	A random permutation scoring as well as the real order means the model uses no spatial structure and the mapping is not meaningful.
Direct recording-identity target	Trains a classifier to predict which of a participant's two recordings a window came	Measures directly how available the identity shortcut is, rather than

	from, which is the medication label up to a flip.	inferring it. Verified in code as identical to the label for 27 of 27 participants.
Relative band power and aperiodic exponent under both normalisations	Repeats the identity probe with features that are scale-free by construction, so channel standardisation cannot change them.	A drop under standardisation would have meant standardisation removes the shortcut. These features show it does not.
Pooling the model across participants	Trains one shared model over all 27 participants instead of one model per participant, so each class draws on 27 recordings rather than two.	This is the design change that closes the identity shortcut, and the channel-amplitude probe is what measures whether it worked.
Cross-participant re-run of the whole protocol	Repeats the analysis with the test participant's recordings absent from training entirely.	Separates an effect shared between people from one that is specific to an individual. A drop bounds the claim; it does not by itself identify the cause.
Pairwise transfer across 702 ordered participant pairs	Applies each participant's model to every other participant, against a null built by refitting every source model over 200 label permutations.	A maximum over 702 pairs is large by construction, so the null of the maximum is what makes the observed maximum interpretable.
Per-window aperiodic fitting	Recomputes the aperiodic exponent for every window rather than once per session.	A session-constant feature takes one value per recording and so is collinear with the label by construction, scoring near-perfectly for reasons unrelated to EEG.

Table 3-3. Control checks, what each one is, and the explanation it rules out. Every control used in the study, paired with the specific alternative explanation for the main result that its outcome eliminates.

Two conventions govern how these controls are read. The first is that a control built on features that cannot carry cortical information sets a floor, not a comparison, the question it answers is whether a score of a given size is available without any brain signal at all, and a trained model that fails to beat such a floor has demonstrated nothing regardless of its absolute value. The second is that positive and negative controls answer different questions and both are needed. The negative control shows the pipeline does not invent signal but rather the positive control shows it can find signal that is present. A study with only the first cannot distinguish a real null from a broken pipeline.

Two of these controls need scoping. The aperiodic control could not reuse the cohort fits computed earlier in the project, which were produced once per session per channel, a session

constant feature is collinear with the medication label by construction, so the fits were recomputed for every window.

The cross-participant re-run of Section 3.6 is the only control in the study that can distinguish an effect shared between people from one specific to a single recording, because every other control operates inside a design where the two possibilities have the same observable consequences.

Table 3-4. Pre-registered thresholds.

Threshold	Value	Fixed at
Trivial floor a model must clear	0.614 balanced accuracy, the duration-only heuristic	Pre-registration, 2026-07-26
Outcome bands	positive > 0.65; weak positive 0.60–0.65; null ≤ 0.60	Pre-registration, 2026-07-26
Primary significance test	refit-based null; post-hoc permutation treated as optimistic	Pre-registration, 2026-07-26
Predicted outcome	null	Pre-registration, 2026-07-26
Staging gate for the alignment experiment	proceed only if the intervention moves the frozen probe by ≥ 0.04 under duration-matched evaluation	Pre-registration, 2026-07-26
Positive-control absolute criterion	identity top-1 > 0.40	Before the input ceiling was measured
Positive-control ratio criterion	exceed the raw band-power ceiling	Same stage; this is the criterion used
Seed-spread gate for attribution	flag at ±0.03, stop at ±0.05	Attribution plan, before retraining
Biomarker certification rule	generalises under leak-free cross-participant task split, confidence interval excluding chance, permutation $p < 0.05$	Attribution plan, 2026-08-07

Table 3-4. Pre-registered thresholds. Every threshold, decision rule and predicted outcome fixed in advance of seeing results, with the document and date each was pre-registered against.

The positive control's absolute threshold was set before the input ceiling had been measured, and it proved mis calibrated, the raw features reached a value that no model could have cleared on merit. The pre-specified ratio criterion is the one applied. Both figures are reported, and the criterion was not switched.

One deviation from the pre-registration is declared. The document specified a final experiment applying Euclidean alignment with session-balanced sampling and duration-matched evaluation, staged behind the gate in Table 3-4. That experiment was not carried out. The leave-one-patient-out arms had by then returned a null across all five configurations, and none had cleared the trivial floor, so an alignment step applied to those arms would have been tuning a procedure that had no measured signal to align. The remaining project time was committed instead to the task-split design of Section 3.6 and the attribution work of Section 3.7, which address the same substantive question under a design the data can support. The floor, the outcome bands, the primary test and the predicted outcome all remain in force.

Significance is assessed against a refit-based null: 100 complete repetitions of the pipeline with the medication labels shuffled within each participant. Shuffling within participant rather than across the cohort preserves the cohort structure, the window counts and the participant groupings, and destroys only the assignment of medication state, so the null measures what the procedure can achieve on this data when the label carries no information. This replaced a post-hoc permutation test, meaning one applied to predictions that had already been computed, without refitting the classifier to the shuffled labels. A post-hoc test cannot capture the classifier's ability to adapt to noise and therefore understates the null. The two disagreed materially on the classical arm, the same observed value of 0.657 gives $p = 0.005$ under the post-hoc test and $p = 0.020$ against the refit-based null, and the refit null's maximum across 100 repetitions was itself 0.657. Post-hoc values are reported for comparability and are labelled as optimistic wherever they appear.

The 0.5 threshold may be miscalibrated for individual participants, so two threshold-free tests supplement the primary metric, an exact binomial sign test on whether each participant's OFF session scores above their own ON session, and a Wilcoxon signed-rank test on the within-participant difference in mean predicted probability. The two disagree on the session-duration asymmetry. A sign test discards the size of each difference, and on this cohort the counts are close to even, whereas where ON sessions are longer, they are longer by more, which is why the signed-rank test reaches $p = 0.044$ and the sign test does not. Session duration is therefore a usable cue within a participant rather than a consistent cohort-wide difference in one direction, and it is a confound for within-participant designs specifically.

Multiplicity is handled by declaration. The pre-registered primary comparison, which is the low-rank adaptation arm under the leave-one-patient-out design of Section 3.6, is a single comparison and is not adjusted. The classical control was run in several declared variants, all of which are reported, and its nominal significance does not survive correction for the number of variants examined. Choosing a layer or a configuration by inspecting held-out performance is selection on the test set, wherever such a figure appears, the selection procedure that produced it appears alongside it.

Invariant checks abort the run instead of emitting a warning. They cover clock alignment, agreement of the results schema, and any window appearing on both sides of a split. Seeds are fixed at ICA fitting, at shuffling and at initialisation, and the cleaned signal is cached before windowing, so any two experiments sharing a configuration consumed byte-identical input.

4. Results

4.1 Realised cohort and data yield

This chapter reports two analysis sets, which are never merged. The main analysis uses the task-split subset of 27 participants. The exploratory work of Section 4.6 uses the wider cohort of 35 participants from which that subset was drawn. Every table and figure below states which set it comes from, and no estimate from one set is compared numerically with an estimate from the other.

The task-split subset comprises 27 participants, those for whom both kitchen tasks, tea and toast, could be recovered with disjoint spans and at least five evaluation windows in both medication blocks. It yields 54 recordings, one per participant per medication state, subdivided into 108 participant-by-task-by-state cells. Under the 5 s training stride the subset yields 9,963 windows; under the 10 s non-overlapping evaluation stride it yields 4,980 windows over the same recording time. The two tilings overlap and are not additive. Both kitchen tasks are logged in both blocks for 34 of the 35 participants in the wider cohort, so task type is not confounded with medication state.

Eight participants in the wider cohort are absent from the subset. Seven were lost to task-recovery failure in one or both blocks, and one to an overlap between the logged start and end times of the two tasks in a single session. A corrupted-timestamp defect in the task-recovery search was found and corrected during this reduction, Chapter 3 describes it.

The wider cohort yields 70 sessions, 35 ON and 35 OFF, and 37,174 windows, of which 20,122 come from the ON block and 17,052 from the OFF block. Every run reported below reproduced these totals, as the invariant fixed in Chapter 3 requires.

ON and OFF sessions differ in length. Across the 35-participant cohort the median non-overlap window count is 288 for ON sessions and 239 for OFF sessions, a paired difference that reaches significance under a Wilcoxon signed-rank test ($p = 0.044$, $n = 35$ pairs). The direction is not uniform, which of a participant's two sessions is the longer varies between participants rather than following a single cohort-wide pattern. A signed-rank test responds to the ranked magnitudes of the paired differences and does not require a uniform sign, so a systematic length difference at cohort level is compatible with participant-level variation in its direction. Section 4.3 reports the duration-matched re-run that addresses this.

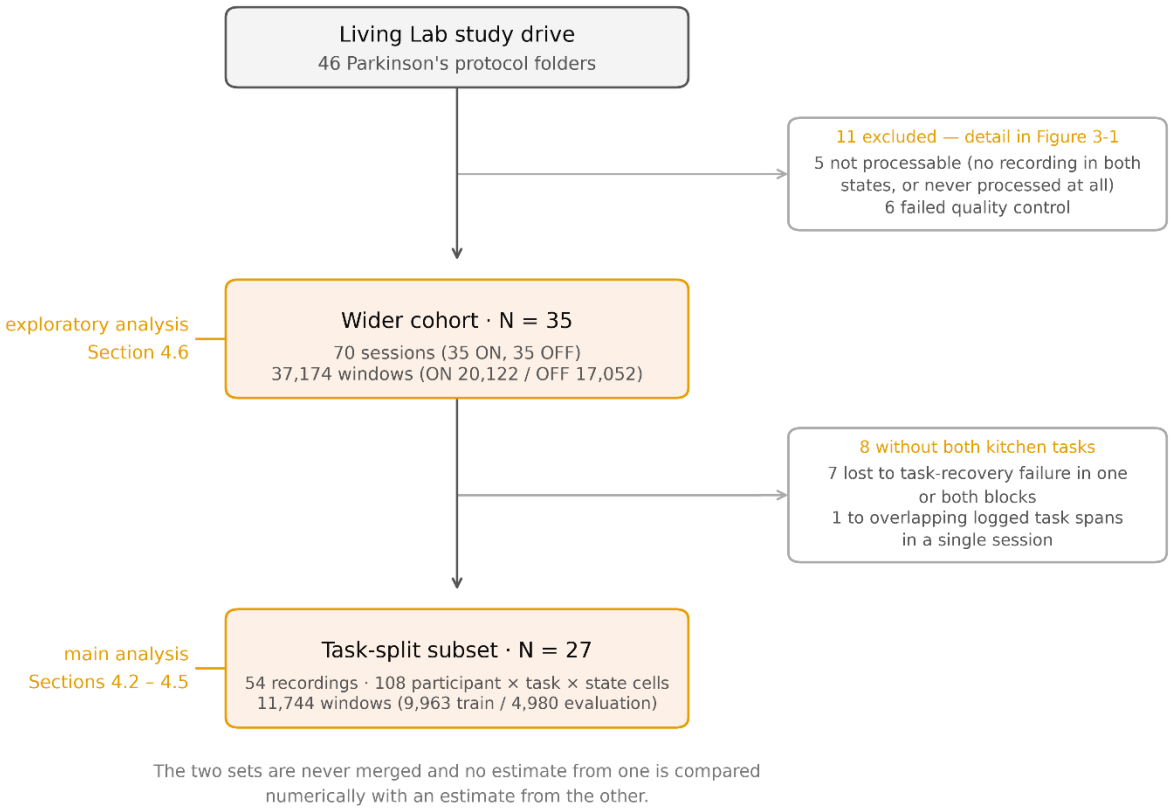


Figure 4-1. Cohort flow from the study drive to the two analysis sets. Eleven of the 46 Parkinson's-protocol folders are excluded before the cohort is fixed, five because no usable recording exists in both medication states or the recording was never processed, and six under the quality-control criteria set in the audit configuration. Figure 3-1 names each exclusion and the stage it belongs to. The wider cohort of 35 participants is the set used for the exploratory leave-one-participant-out work of Section 4.6. A further eight participants are absent from the task-split subset, seven because task recovery failed in one or both blocks and one because the logged start and end times of the two tasks overlapped in a single session. The remaining 27 participants form the task-split subset that carries the main analysis of Sections 4.2 to 4.5. The two sets are never merged, and no estimate from one is compared numerically with an estimate from the other.

4.2 Main analysis, cross-task decoding of medication state

Analysis set, task-split subset, N = 27 participants, 54 recordings. One shared model is trained on windows from one kitchen task and tested on the other.

The main analysis asks whether a model trained on one kitchen task can identify medication state in windows from the other (Section 3.6). Both directions are reported separately, toast-to-tea, in which the model trains on toast windows and is tested on tea windows, and tea-to-toast, its reverse. Balanced accuracy is the primary metric and chance is 0.5 by construction.

An earlier per-participant variant of this design was abandoned on the evidence of its own control. The channel-amplitude probe, built on a feature with no capacity to carry cortical information, reached a participant-mean balanced accuracy of 0.934 under that variant and exceeded every trained arm, confirming the recording-identity route set out in Section 3.6. The pooled design reported here was adopted in response, and Section 4.3 reports what happened to the amplitude control under that change.

Table 4-1 gives every arm in both directions, duration-matched and unmatched. Figure 4-2 shows the same estimates against the amplitude control.

Table 4-1. Pooled task-split results by arm and direction (task-split subset, N = 27, channel-standardised input, participant-mean balanced accuracy with 95% cluster-bootstrap confidence intervals over participants)

Arm	Toast-to-tea, matched	95% CI	Tea-to-toast, matched	95% CI	Toast-to-tea, unmatched	Tea-to-toast, unmatched
Full fine-tuning	0.832	[0.792, 0.869]	0.710	[0.667, 0.752]	0.844	0.701
LoRA	0.694	[0.654, 0.735]	0.610	[0.554, 0.665]	0.717	0.626
Frozen probe	0.597	[0.551, 0.647]	0.595	[0.545, 0.645]	0.608	0.599
Channel-amplitude control	0.525	[0.501, 0.549]	0.535	[0.503, 0.565]	0.538	0.536

Table 4-1. Pooled task-split results by arm and direction (task-split subset, N = 27, channel-standardised input, participant-mean balanced accuracy with 95% cluster-bootstrap confidence intervals over participants). Balanced accuracy for each adaptation arm and the channel-amplitude control, reported separately for the toast-to-tea and tea-to-toast directions and for the duration-matched and unmatched evaluation conditions.

Full fine-tuning of all approximately 15.2 million backbone parameters gives the highest estimate in the study, 0.832 (95% CI [0.792, 0.869]) in the toast-to-tea direction under duration matching. The same arm gives 0.710 (95% CI [0.667, 0.752]) in the tea-to-toast direction. Both intervals lie well clear of chance and well clear of the channel-amplitude control. LoRA reaches 0.694 and 0.610 in the two directions, and the frozen probe, which trains no backbone parameters, reaches 0.597 and 0.595. Adaptation capacity therefore orders the arms in both directions, the more of the backbone that is allowed to move, the higher the estimate.

The gap between the two directions is substantial and it is a property of the result rather than a nuisance. Toast-to-tea exceeds tea-to-toast by 0.122 under full fine-tuning and by 0.084 under LoRA, while the two frozen-probe estimates differ by 0.002. Section 4.5 returns to this asymmetry, since the band-stop sweep finds the two directions depend on different parts of the spectrum.

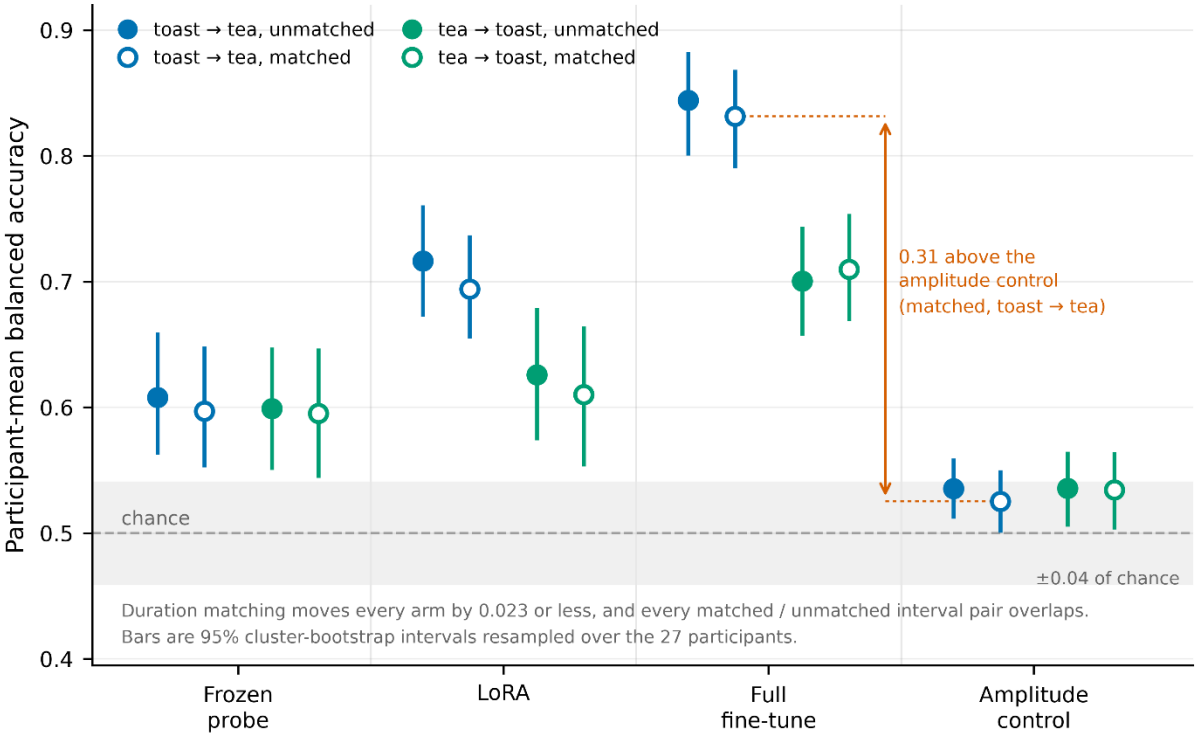


Figure 4-2. Pooled task-split decoding of medication state in both directions (task-split subset, N = 27). Participant-mean balanced accuracy for each arm, shown unmatched (filled) and duration-matched (open), with 95% cluster-bootstrap confidence intervals resampled over participants. The channel-amplitude control, built on a feature with no capacity to carry cortical information, sits within 0.04 of chance in both directions and in both matching conditions. Full fine-tuning exceeds it by 0.31 balanced-accuracy points in the toast-to-tea direction. Duration matching moves every arm by 0.023 or less and every pair of intervals overlaps.

4.3 Controls on the main analysis

Analysis set, task-split subset, N = 27. Controls run under the identical protocol to Section 4.2 unless stated.

Three alternative explanations for the Section 4.2 result were tested directly, that the model recognises individual recordings, that it exploits the difference in session length, and that the pipeline manufactures signal from noise.

The recording-identity explanation is the one the per-participant design failed. Under pooling, the channel-amplitude control gives 0.525 (95% CI [0.501, 0.549]) in the toast-to-tea direction and 0.535 (95% CI [0.503, 0.565]) in the tea-to-toast direction. Both intervals sit at or barely above chance, against 0.832 and 0.710 for full fine-tuning on identical data under an identical protocol. The margin is 0.307 and 0.175 respectively. Per-channel amplitude is the most direct expression of an electrode-contact fingerprint available from these recordings, and it reached 0.934 when the design permitted that route. Its collapse to chance under pooling is measured evidence that the route is closed, rather than an argument that it should be.

Channel-wise standardisation is applied to the input in every arm reported here, which removes the per-channel amplitude scale before the model sees the data. That transform alone does not close the identity route in a within-participant design, a classifier trained directly to predict which of a participant's two recordings a window came from scores 0.963 on raw input and 0.924 under channel standardisation. The closure reported above comes from pooling rather than from standardisation.

Duration matching, specified in Section 3.6, addresses the session-length difference reported in Section 4.1. It cost 10.0% of training windows in the toast-to-tea direction and 11.2% in the tea-to-toast direction, moved every arm by 0.023 or less, and left every matched and unmatched confidence interval overlapping. Session length therefore does not account for the result, and the duration-matched estimates are carried forward.

The negative control repeated the full pipeline 100 times with medication labels shuffled within participant, refitting the classifier on each repetition. The resulting null distribution was centred at 0.500 with a standard deviation of 0.058. No window, overlapping or otherwise, crossed a train-test boundary at any level of the design, and shuffled-label performance showed no elevation above chance.

Two further probes bear on what survives standardisation. Relative band power is scale-free by construction and is unchanged by channel standardisation, moving from 0.910 to 0.911 within participant, the aperiodic exponent remains at 0.886. Neither figure is a result of the main analysis. Both establish that scale-free features retain a within-participant route that amplitude-based standardisation cannot remove, which is why the pooled design rather than the normalisation scheme carries the argument.

4.4 Transfer to unseen participants

Analysis set, task-split subset, N = 27, 108 participant × task × state cells. The same task-split protocol is re-run with the test participant's recordings absent from training.

Sections 4.2 and 4.3 establish that the pooled model separates medication state across a change of behavioural task, and that neither recording identity nor session length accounts for the separation. A further question is whether the learned effect is shared between individuals. The test is a re-run of the identical protocol with the test participant's recordings removed from training altogether, other participants' windows of one task for training, the held-out participant's windows of the other task for evaluation.

Under this design, CodeBrain's arms return to chance. Pooled over both task directions, the frozen probe gives 0.519 (95% CI [0.444, 0.602], p = 0.363), LoRA 0.426 (p = 0.955) and full fine-tuning 0.435 (p = 0.945). Under channel-wise standardisation the two trained arms recover slightly, LoRA to 0.556 (p = 0.139) and full fine-tuning to 0.537 (p = 0.199), without either reaching significance. The channel-amplitude control gives 0.472, with 5 of 27 participants above chance.

A pairwise transfer test asked whether any individual participant's model generalised to any other participant. Each of the 27 × 27 ordered pairs were evaluated, giving 702 off-diagonal cells in which a source participant's model was applied to a different target participants held-out-task window.

The observed maximum off-diagonal score was 0.984 on raw features and 0.856 under channel standardisation. Each was compared against a null distribution of the maximum, built by permuting labels and refitting every source model over 200 permutations. Both observed maxima sat inside their respective nulls (null mean 0.980 ± 0.010, p = 0.736; null mean 0.902 ± 0.022, p = 0.965), and the off-diagonal mean was indistinguishable from chance under both schemes, at 0.496 and 0.500.

The medication-state effect the pooled model reads therefore does not transfer between individuals at this sample size. A transfer test alone does not establish why, since a genuine but individually expressed drug response and a participant-specific artefact produce the same pattern under this design, Chapter 5 takes up the distinction. No estimate in this chapter supports deployment to a patient the model has never seen.

One feature family does transfer. Relative band power gives 0.620 (95% CI [0.519, 0.713], AUC 0.645) pooled over both directions, identically under both normalisation schemes, and 0.685 (95% CI [0.574, 0.778]) in the toast-to-tea direction alone. This figure carries a post-hoc permutation p-value of 0.005 and no refit-based one, and it inherits the duration confound of Section 4.1 unchanged, since this run was not duration-matched. The channel-amplitude control under channel-wise standardisation also shows a small residual under this design (0.583, 95% CI [0.509, 0.657], p = 0.020), which is one result among eight controls tested at the 0.05 level without correction for multiplicity. Section 4.6 reports the corresponding classical estimate on the wider cohort, and Chapter 5 treats the two together as the study's single transferable candidate.

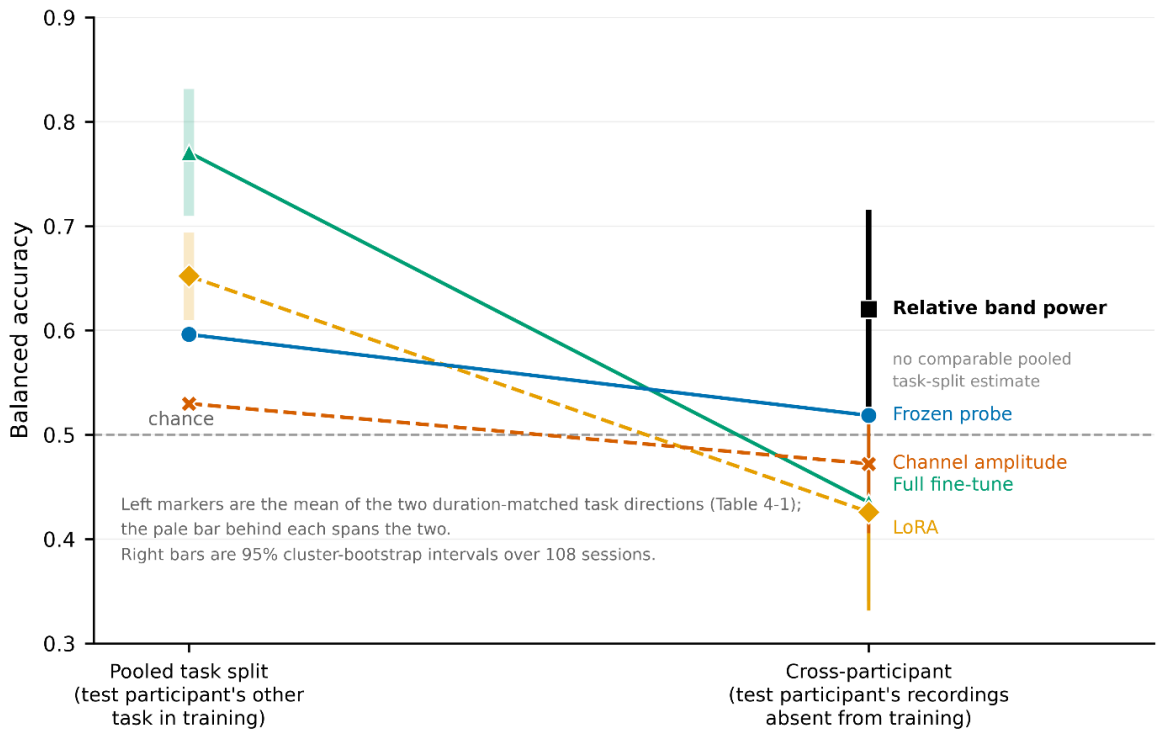


Figure 4-3. Arms and controls under a change of split (task-split subset, N = 27). Each arm is evaluated twice under an identical protocol, once pooled with the test participant's other recording present in training and once with it absent, and the two estimates are joined. Left-hand markers are the mean of the two duration-matched task directions of Table 4-1, with the pale bar behind each spanning the two directions; right-hand markers are pooled over both directions, with 95% cluster-bootstrap intervals over 108 participant × task × state cells.

CodeBrain's arms return to chance when the test participant is removed from training, and so does the channel-amplitude control. Relative band power is the only feature family that does not, retaining 0.620 (95% CI [0.519, 0.713]) across participants; it is shown at its cross-participant value alone, because it has no comparable pooled task-split estimate of medication state.

4.5 Attribution and spectral dependence

Analysis set, task-split subset, N = 27, pooled duration-matched full fine-tuning arm, the highest-scoring model in the study (Section 4.2). Five retraining seeds.

Attribution was applied to the arm that scores highest in the main analysis, retrained under five seeds as specified in Section 3.7.

Balanced accuracy across the five seeds was 0.841, 0.810, 0.836, 0.810 and 0.835, a mean of 0.826 with a spread of 0.031. The pre-registered stability gate flags at ±0.03 and stops at ±0.05. The spread crossed the flag threshold without reaching the stop threshold, so the analysis proceeded and every figure in this section carries that flag.

Two candidate readouts were tested and both were abandoned on measurement grounds. A readout of the classification head's learned filter response was indistinguishable from its response at random initialisation, with a peak-to-trough ratio of 1.41 to 1.47 when trained against a null of 1.42 whose 95th percentile was 1.56. The head does not adapt materially during fine-tuning, and adaptation is concentrated in the backbone. The pretrained vector-quantised codebooks were confirmed unused in the downstream forward pass, their weights byte-identical to the pretrained checkpoint after fine-tuning, so no codebook-based readout can account for this model.

Phase randomisation, which preserves the amplitude spectrum while destroying waveform structure, cost 0.051 (± 0.021) balanced accuracy. That is a small effect and it follows from the architecture, since the patch embedding computes the magnitude of a real fast Fourier transform and discards phase. One-second patch occlusion cost between 0.007 and 0.025, with no position-dependent pattern. The model therefore reads a distributed amplitude-spectrum signature rather than a localised waveform event.

A band-stop sweep of 469 bands, constructed as described in Section 3.7, characterised the frequency content that signature depends on. An initial version using 2 Hz notches found no dependence on any band, that null was withdrawn, since such a notch removes only about 4% of the power between 1 and 45 Hz and the result described the notch rather than the model.

Every band reduced balanced accuracy. None of the 469 improved it, and the reductions ranged from 0.0003 to 0.1707. The most damaging band, 1 to 19 Hz, removed 98.6% of the power between 1 and 45 Hz and reduced balanced accuracy from 0.826 to 0.655. Per unit of power removed, bands in the 23 to 41 Hz range were 11.8 times more damaging than the sweep average.

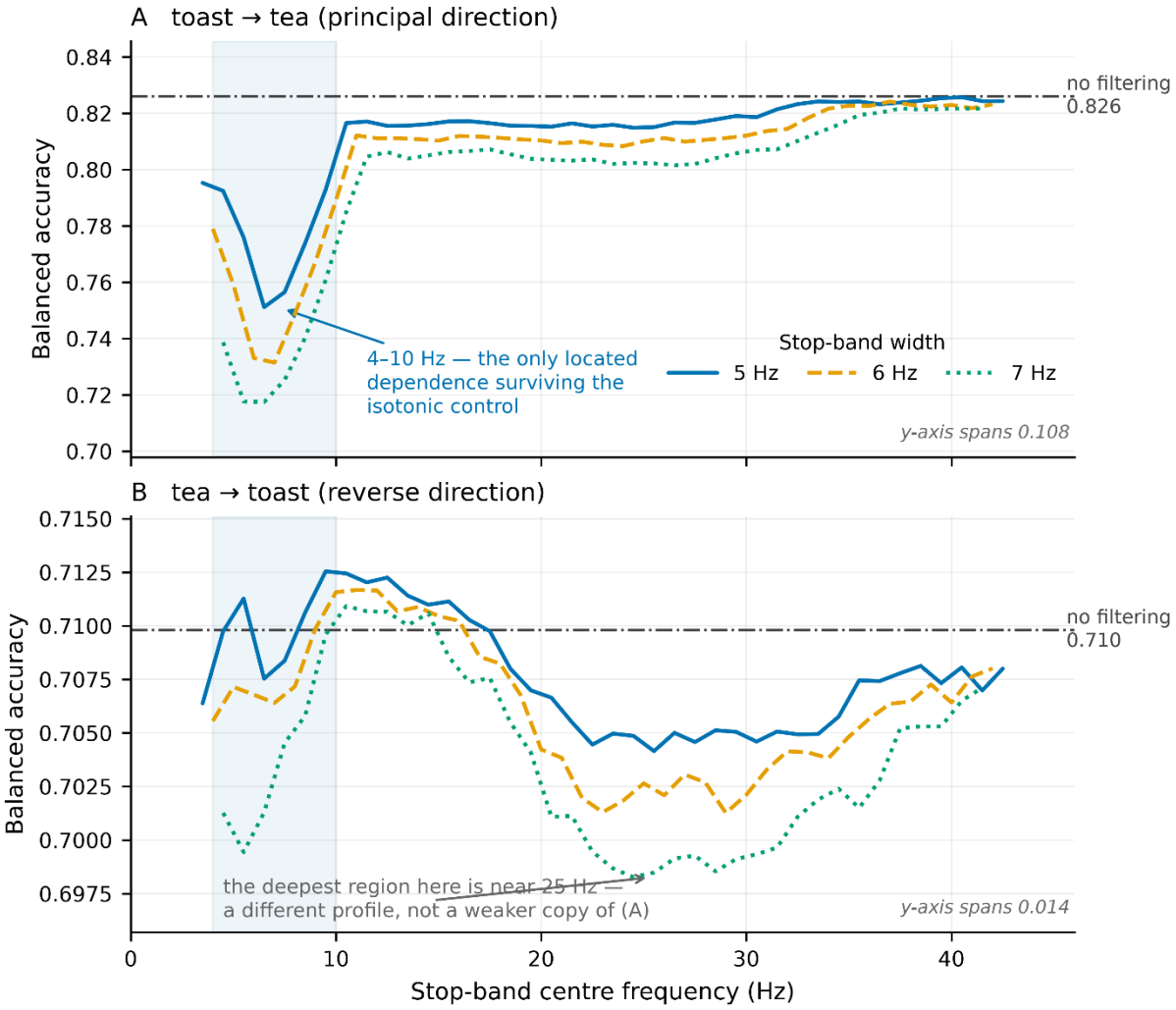


Figure 4-4. Band-stop sweep over the pooled duration-matched full fine-tuning arm (task-split subset, N = 27), averaged over five retraining seeds. Balanced accuracy after removal of a band of the stated centre frequency, for (A) the toast-to-tea direction and (B) the tea-to-toast direction, with each direction's unfiltered baseline marked. One curve is drawn per stop-band width; only widths of 5 to 7 Hz are shown, because at 8 Hz and above the best band sits on the sweep's lower boundary and is not interpretable as a located effect. The two panels carry independent y-axes and very different ranges — 0.108 in (A) against 0.014 in (B) — so curve heights are comparable within a panel but not between them; each panel is annotated with its own span. In the principal direction all 469 bands reduced balanced accuracy; in the reverse direction 23 raised it, by no more than 0.0027 and with none agreed on by all five seeds. The located 4 to 10 Hz dependence in (A) is not reproduced in (B), whose deepest region lies near 25 Hz instead. Results in this section carry the seed-stability flag described in the text, the observed max–min spread of 0.033 having crossed the pre-registered flag threshold of 0.03 without reaching the stop threshold of 0.05.

The two within-width controls specified in Section 3.7, a within-width standardisation and a per-width isotonic regression against the fraction of power removed, were applied before the sweep was interpreted.

After controlling for the fraction of power removed, one position-specific effect survives. It is an interior optimum at the narrow end, centred on 4 to 10 Hz, with a residual of +0.023 at 5 Hz width, present in five of five seeds. At widths of 8 Hz and above the best band sits on the sweep's lower boundary, so those edges record where the search stopped and only the narrower widths can be interpreted.

The same sweep run in the tea-to-toast direction, with a baseline of 0.710, does not reproduce this profile. The interior effect at widths of 5 to 7 Hz is 8 to 12 times smaller, at 0.006 to

0.010 against 0.075 to 0.109, and the marginal profile peaks near 25 Hz instead of decaying from 1 Hz. The two directions therefore depend on different regions of the spectrum, which is consistent with the accuracy asymmetry reported in Section 4.2 and which no single-direction analysis would have revealed.

No candidate from the attribution analysis was submitted to the pre-registered biomarker certification gate, since certification requires generalisation to unseen participants and Section 4.4 shows that the fine-tuned model does not provide it.

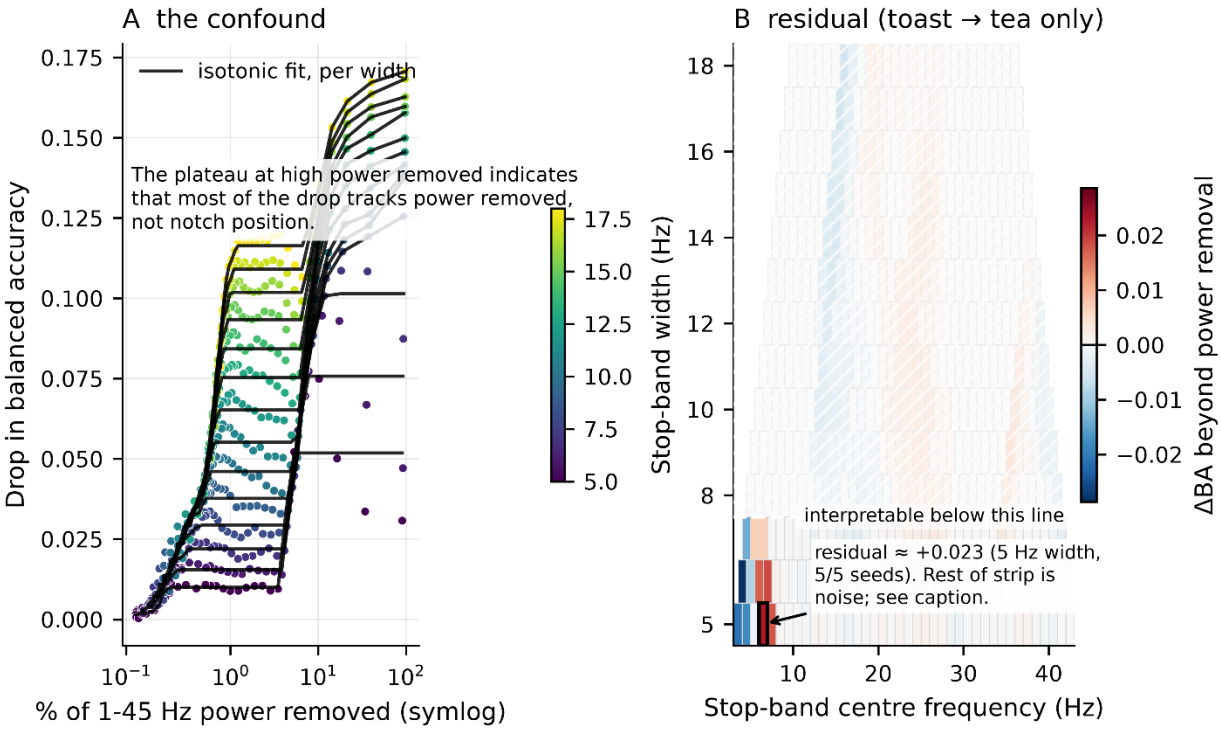


Figure 4-5. Separation of the band-stop effect from the quantity of power removed (toast-to-tea direction, five seeds). (A) Drop in balanced accuracy (BA) against the percentage of 1 to 45 Hz power removed by each band, coloured by stop-band width, with an isotonic fit per width. The plateau at high power removed indicates that most of the drop tracks the quantity of power removed rather than the position of the band. (B) Residual change in balanced accuracy once the relationship in (A) has been accounted for. The only position-specific effect surviving this control is an interior optimum at the narrow end, approximately +0.023 at 5 Hz width, centred on 4 to 10 Hz and present in five of five seeds. The remainder of the interpretable strip is consistent with noise, and cells at widths of 8 Hz and above are not interpretable as located effects.

4.6 Earlier exploratory work, leave-one-participant-out decoding

Analysis set, wider cohort, N = 35 participants, 70 sessions, session-level leave-one-participant-out cross-validation. Not comparable with Sections 4.2 to 4.5.

Before the task-split design was adopted, the project evaluated a session-level leave-one-participant-out design on the wider cohort of 35 participants. That work is reported here in summary because it fixed the pre-registered thresholds against which Section 4.7 reports, and because its classical arm supplies the transferable candidate discussed in Section 4.4. It is exploratory relative to the main analysis and its estimates answer a different question, namely whether medication state can be read from a participant whose recordings the model has never seen.

Five configurations of CodeBrain were evaluated. The frozen probe, mean-pooled over 200 features, gave 0.443 (95% CI [0.357, 0.529]), a flattened alternative carrying 36,000 features gave 0.457 (95% CI [0.371, 0.543]). Low-rank adaptation gave 0.557 (95% CI [0.457, 0.643]), the highest of the five and the only one whose interval lay mostly above 0.50. Full fine-tuning returned 0.443 and LoRA with per-window global standardisation returned 0.443. All five fall inside the pre-registered null band of 0.60 or below, and every interval contains chance. Significance for these arms rests on a post-hoc permutation test, which the pre-registration treats as optimistic, under it, no arm reached $p < 0.05$.

A trivial floor was pre-registered before any modelling result had been examined, a classifier given only each session's non-overlap window count, with no EEG signal, evaluated under the same design. It scored 0.614. None of the five configurations exceeded it. A related heuristic using only each window's elapsed time within the session scored 0.571. These two figures are the reason session length was treated as a first order confound throughout, and the reason the main analysis is duration matched.

Two controls established that the null was a property of the data rather than of the pipeline. The negative control of Section 4.3 was computed under this design and centred at 0.500. A positive control decoded participant identity from the same session-level embeddings, training on ON-block windows and testing on OFF-block windows. Raw band-power features supported 35-way identification at top 1, 0.328, which is the ceiling the input can supply. CodeBrain's final-layer representation gave 0.396, a ratio of 1.21 over that ceiling, and a sweep across all eight residual blocks ranged up to 0.442 at block 4. The pre-registered absolute criterion of top $1 > 0.40$ was fixed before the ceiling had been measured and the final layer does not clear it, the pre-registered ratio criterion is cleared and is the one applied. The representation encodes which participant a recording came from far more strongly than which medication state that participant was in. A third control rejected the electrode-to-channel mapping as an explanation, the study's own 10-20 montage order matched or exceeded the order reported in CodeBrain's downstream configuration, and both exceeded a random permutation, which degraded with depth.

A classical spectral baseline was evaluated under the identical design. Six feature-set variants were declared in advance. The beta-only variants sat at chance, at 0.486 under both normalisations, so the a priori beta-band hypothesis is not supported on these recordings. Relative power gave 0.643, rising to 0.657 once the gamma band was removed under a leave-one-band-out check. That 0.657 was chosen by inspecting held-out performance, so it is reported alongside its correction rather than in place of it, Bonferroni correction across the six declared variants raises the threshold to $p < 0.0083$, and the refit-based p of 0.020 does not clear it. A nested procedure removed the selection bias directly, selecting among candidate feature sets within each outer fold using only the 34 training participants, and gave 0.629 (95% CI [0.543, 0.714], 24 of 35 participants correctly ordered). Inner selection was stable, choosing relative power without gamma in 19 of 35 folds and relative power in a further 15.

A narrow-band sweep across 0.5 Hz bins located the largest within-participant OFF-minus-ON elevation between 4.0 and 5.5 Hz, broad and similar in magnitude at lateral and midline channels. The best narrow band was 2 to 4 Hz at 0.614, below the 4 to 6 Hz range classically associated with parkinsonian rest tremor, which itself scored 0.571. At this sample size the individual bands are not statistically distinguishable from one another and no single band is

reported as a discrete finding. Combining band-power features with CodeBrain's frozen embedding did not improve on the classical features alone under any concatenation scheme tested, giving 0.586, 0.557 and 0.557 against 0.657.

Duration-matched re-runs of the classical arm, taking equal numbers of windows from either the start or the middle of each session, scored between 0.543 and 0.643 across six combinations. Several fall at or below the refit-based null's 95th percentile of 0.600.

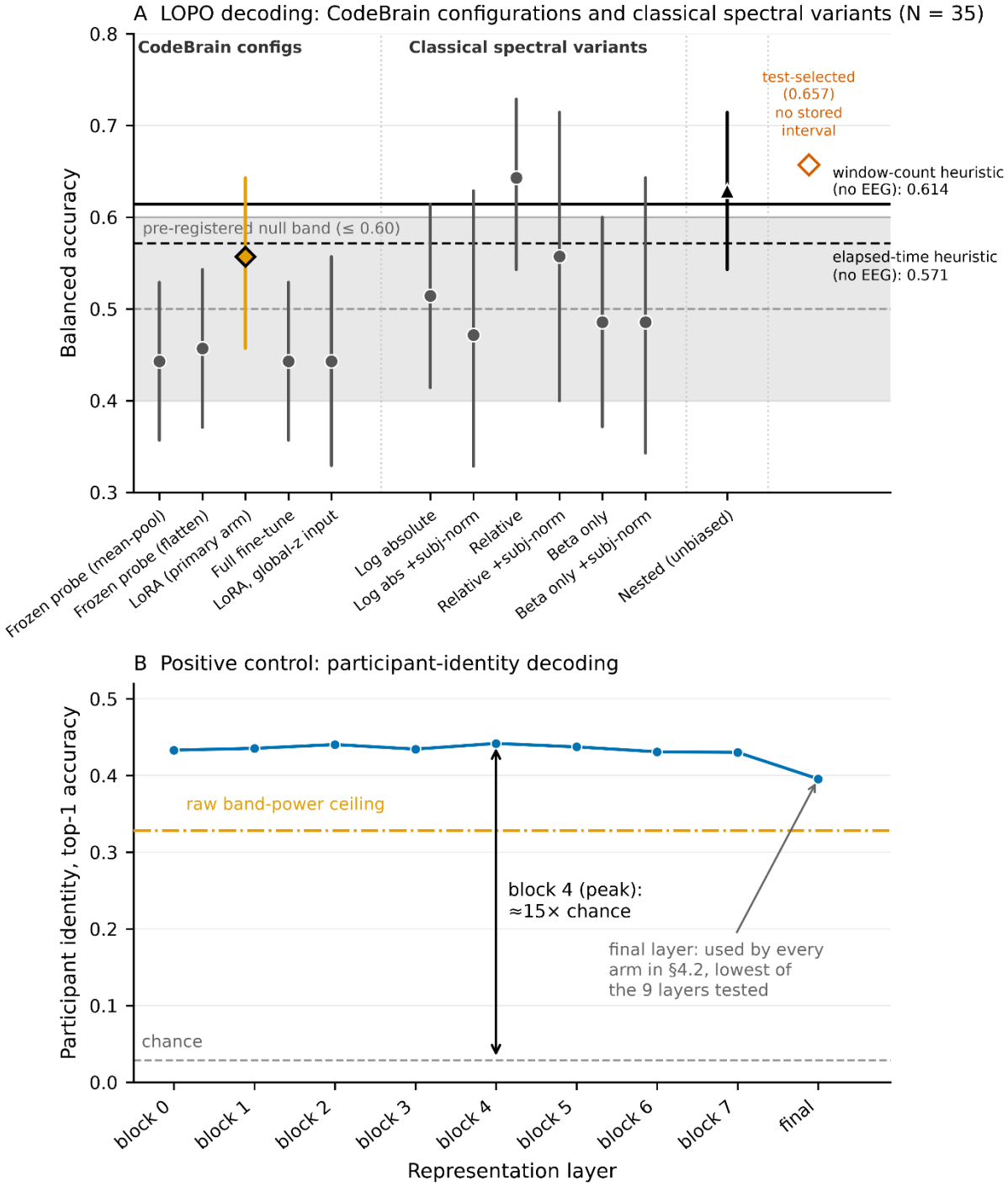


Figure 4-6. Earlier exploratory work, leave-one-participant-out decoding on the wider cohort (N = 35 participants, 70 sessions). (A) Balanced accuracy for five CodeBrain configurations and for the classical spectral variants, with 95% cluster-

bootstrap confidence intervals resampled over participants. The shaded region marks the pre-registered null band (≤ 0.60), and the horizontal lines mark the window-count floor (0.614) and the elapsed-time heuristic (0.571), neither of which uses any EEG signal. (B) Participant-identity decoding from each residual block and from the final layer, against the ceiling supported by the raw band-power input (0.328) and chance (0.029). These estimates are not comparable with Sections 4.2 to 4.5.

4.7 Summary against the pre-registered thresholds

Table 4-2 lists every arm in the study with its analysis set, and Table 4-3 compares the pre-registered thresholds fixed in Chapter 3 against their observed values.

Table 4-2. Master results table.

arm	analysis set	unit / n	balanced accuracy	95% CI	AUC	p	outcome
Full fine-tune, toast-to-tea, matched	task-split pooled, N=27	27 participants	0.832	[0.792, 0.869]	n/a	n/a	main result
Full fine-tune, tea-to-toast, matched	task-split pooled, N=27	27 participants	0.710	[0.667, 0.752]	n/a	n/a	main result, reverse direction
Full fine-tune, toast-to-tea, unmatched	task-split pooled, N=27	27 participants	0.844	[0.800, 0.883]	n/a	n/a	main result, unmatched
LoRA, toast-to-tea, matched	task-split pooled, N=27	27 participants	0.694	[0.654, 0.735]	n/a	n/a	main analysis
LoRA, tea-to-toast, matched	task-split pooled, N=27	27 participants	0.610	[0.554, 0.665]	n/a	n/a	main analysis
Frozen probe, toast-to-tea, matched	task-split pooled, N=27	27 participants	0.597	[0.551, 0.647]	n/a	n/a	main analysis
Frozen probe, tea-to-toast, matched	task-split pooled, N=27	27 participants	0.595	[0.545, 0.645]	n/a	n/a	main analysis
Channel-amplitude control,	task-split pooled, N=27	27 participants	0.525	[0.501, 0.549]	n/a	n/a	control, at chance

toast-to-tea							
Channel-amplitude control, tea-to-toast	task-split pooled, N=27	27 participants	0.535	[0.503, 0.565]	n/a	n/a	control, at chance
Channel-amplitude, per-participant design	task-split per-participant, N=27	54 records	0.934	[0.901, 0.961]	n/a	n/a	control that failed; design abandoned
Cross-participant frozen probe	task-split, N=27, cross-participant	108 participant × task × state cells	0.519	[0.444, 0.602]	0.484	0.363 (post-hoc)	no transfer
Cross-participant LoRA	task-split, N=27, cross-participant	108 participant × task × state cells	0.426	[0.333, 0.519]	0.401	0.955 (post-hoc)	no transfer
Cross-participant full fine-tune	task-split, N=27, cross-participant	108 participant × task × state cells	0.435	[0.343, 0.528]	0.413	0.945 (post-hoc)	no transfer
Cross-participant relative band power	task-split, N=27, cross-participant	108 participant × task × state cells	0.620	[0.519, 0.713]	0.645	0.005 (post-hoc; no refit)	transferable candidate
Frozen probe (mean-pool)	exploratory, LOPO, N=35	70 sessions	0.443	[0.357, 0.529]	0.482	0.950 (post-hoc)	null
Frozen probe (flatten)	exploratory, LOPO, N=35	70 sessions	0.457	[0.371, 0.543]	0.467	0.841 (post-hoc)	null
LoRA	exploratory, LOPO, N=35	70 sessions	0.557	[0.457, 0.643]	0.541	0.194 (post-hoc)	null
Full fine-tune	exploratory, LOPO, N=35	70 sessions	0.443	[0.357, 0.529]	0.423	0.945 (post-hoc)	null
LoRA, global-z input	exploratory, LOPO, N=35	70 sessions	0.443	[0.329, 0.557]	0.424	0.871 (post-hoc)	null
Duration-only heuristic	exploratory, LOPO, N=35	70 sessions	0.614	n/a	0.628	n/a	trivial floor

Relative band power, test-selected	exploratory, LOPO, N=35	70 sessions	0.657	n/a (no stored interval)	n/a (not stored)	0.005 post-hoc / 0.020 refit	null after correction
Relative band power, nested (unbiased)	exploratory, LOPO, N=35	70 sessions	0.629	[0.543, 0.714]	0.620	0.005 (post-hoc; no refit computed)	weak positive
Identity control, final layer	exploratory, cross-session identity	35-way	0.396 top 1	n/a	n/a	n/a	positive control

Table 4-2. Master results table. Balanced accuracy, 95% confidence interval, AUC and significance for every arm and control evaluated in the study, organised by analysis set, the main task-split analysis, cross-participant re-runs, the exploratory leave-one-participant-out design, and the identity controls with each result's outcome classification against the pre-registered criteria of Table 3-4.

Table 4-3. Pre-registered thresholds against observed values.

threshold	fixed value	observed	criterion met
Trivial floor	0.614 (duration heuristic)	main analysis 0.832; exploratory LOPO best arm 0.557	cleared by the main analysis, not by LOPO
Outcome bands	positive >0.65; weak 0.60-0.65; null <=0.60	main analysis 0.832 and 0.710	positive in both directions
Primary significance test	refit-based null	exploratory LOPO arm p = 0.194 (post-hoc)	not significant on the exploratory arm
Predicted outcome	null	null on the exploratory LOPO design; positive on the task split	prediction held for the design it was fixed for
Positive-control absolute criterion	top 1 > 0.40	0.396 (final layer) / 0.442 (selected layer)	not met (final layer) / met (selected layer)
Positive-control ratio criterion (criterion used)	exceed raw ceiling 0.328	0.396 or 0.442, both exceed	met
Seed-stability gate	flag +/-0.03, stop +/-0.05	0.033	flag reached, stop not reached
Biomarker certification rule	leak-free generalisation, CI excludes chance, permutation p<0.05	relative band power 0.620, CI [0.519, 0.713], post-hoc p = 0.005	partially met; no refit-based null computed
Staging gate for alignment experiment	EA moves frozen probe >=0.04	experiment not carried out	not evaluated

Table 4-3. Pre-registered thresholds against observed values. Each threshold and decision rule fixed in Table 3-4, set against the value observed once results were available, with a statement of whether the criterion was met.

The pre-registration was written against the leave-one-participant-out design, and its predicted outcome, stated in advance as null, held for that design. The main analysis reported in this chapter uses a different design and clears the trivial floor and the positive outcome band in both directions. The Euclidean alignment experiment staged behind the gate in Table 4-3 was not carried out, the deviation and the reasoning behind it are recorded in Section 3.8.

5. Discussion

5.1 Principal findings

Three findings carry this chapter. Medication state was separable within participants across a change of behavioural task, at 0.832 balanced accuracy against a control at 0.525. That separation did not survive removal of the test participant's own recordings from training. The frequency content the model depended on was distributed rather than localised, with a single position-specific dependence between 4 and 10 Hz surviving in one direction of the split. Together these establish that information about medication state is present in ambulatory dry-electrode EEG, that it did not generalise beyond the individual at this sample size, and that no candidate meets the certification criterion fixed in Chapter 3.

5.2 What the main analysis establishes, and the limit of the claim

The main analysis produced the highest medication-state estimate in the study, 0.832 (95% CI [0.792, 0.869]) under full fine-tuning in the toast-to-tea direction, an interval lying clear of chance and clear of the pre-registered positive outcome band. Three controls reported in Section 4.3 make that separation difficult to dismiss as an artefact of the design, the channel-amplitude probe sat at 0.525 under the identical protocol, duration matching moved every arm by 0.023 or less, and the refit-based negative control was centred at 0.500. Neither an electrode-contact fingerprint, the session-length asymmetry, nor signal manufactured by the pipeline accounts for the result.

The ordering of the three adaptation arms is itself informative. The frozen probe, which trains no backbone parameter, reached 0.597; low-rank adaptation, at roughly 0.6% of the backbone, reached 0.694; full fine-tuning, at 15.2 million parameters, reached 0.832. Accuracy therefore rose monotonically with adaptation capacity. The implication is that the pretrained representation does not contain a medication-state axis in a form a linear readout can exploit. That axis has to be constructed by moving the backbone, which is consistent with the domain gap set out in Chapter 3, the pretraining corpus consists of seated clinical recordings made with wet electrodes against conventional references, whereas these are ambulatory, dry electrode, referenced to linked ears and drawn from a Parkinson's population. Under-adaptation can accordingly be excluded as an explanation for the weaker results elsewhere, since full parameter access was available and was used.

Pooling closed the route by which a single fixed property of one recording could answer the question, and Section 4.3 measured that closure directly, the amplitude control falling from 0.934 to 0.525. What pooling does not close is a participant-conditional solution. Every

participant appears on both sides of the main split, and a backbone with 15.2 million free parameters is capable of learning which participant a window came from and applying that participant's own ON-versus-OFF offset. The positive control establishes that the route is open, participant identity was decodable from the same representation at a top 1 rate of 0.396, against a raw-input ceiling of 0.328 and a chance rate of 0.029. Additionally the identity control was run on the wider cohort. It does not establish that the route was taken, and the capacity ordering above is consistent with either reading, since a drug response expressed differently in each patient would also require free parameters the frozen probe does not have.

What the pooled estimate cannot be is evidence of a marker held in common across patients. It is better read as an upper bound on what per-participant calibration might achieve on these recordings.

A further consequence follows, and the transfer test of Section 4.4 cannot resolve it. A drug response that is genuine but expressed differently in each patient, and a participant-specific nuisance that happens to be stable across a change of task, produce the same observable pattern under this design. Section 5.5 leans towards the former without settling it, a fixed per-participant nuisance would not obviously produce a located 4 to 10 Hz dependence that an independent classical analysis also identifies. Distinguishing them requires evidence the study does not have, repeated visits, concurrent symptom measurement, or an independent measure of the nuisance.

5.3 The direction asymmetry

The two directions of the task split did not agree, and the disagreement scales with adaptation capacity. Toast-to-tea exceeded tea-to-toast by 0.122 under full fine-tuning and by 0.084 under low-rank adaptation, while the two frozen-probe estimates differed by 0.002. The band-stop sweep showed that the disagreement is spectral as well as numerical, the interior optimum between 4 and 10 Hz that survives the isotonic control in the toast-to-tea direction is 8 to 12 times smaller in the reverse direction, whose marginal profile instead peaks near 25 Hz.

A medication effect stable across behavioural context should be approximately symmetric under a reversal of which task carries the training label. The observed asymmetry indicates that a substantial part of the solution is specific to the task the model was trained on. Two candidate sources cannot be separated here. The kettle and toast tasks differ in motor content, and in position within the block, since tea preceded toast in 61 of the 62 sessions from which task spans were recovered. Position within the block carries elapsed time, contact drift and fatigue, all of which move in the same direction as medication state under the fixed ordering described in Section 3.1. The task split removes elapsed time from the split itself, but not from the design.

Two practical conclusions follow. Reporting one direction alone would have overstated the result by 0.122 balanced-accuracy points and concealed a spectral profile that does not replicate, which is why both directions are reported separately rather than averaged. Any future design of this kind should treat train-test direction as a factor to be reported rather than a nuisance to be collapsed.

5.4 Why the effect did not transfer between participants

Removing the test participant's recordings from training returned every CodeBrain arm to chance, with no estimate reaching significance under either normalisation scheme, and the pairwise test across 702 ordered pairs found no source model that generalised to another individual (Section 4.4).

Chapter 2 supplies the reason this outcome should not be surprising. Parkinson's disease is biologically and clinically heterogeneous, with subtypes progressing at different rates and responding unevenly to the same therapy (De Pablo-Fernández et al., 2019; Dadu et al., 2022). The incumbent marker is not universally present even when measured invasively. Darcy et al. (2022) detected beta peaks in 92% of patients off medication but in only 84% of individual hemispheres, and Stanslaski et al. (2024), screening under clinically realistic conditions, identified a usable signal in 84.8% of patients but in both hemispheres in only 65%, falling to 71.4% when an automated algorithm rather than an expert reader performed detection. Demanding that a scalp-recorded marker be shared across 27 unselected participants is a stronger requirement than the invasive literature itself meets, and failing it is weak evidence about the existence of a physiological effect.

The positive control locates the difficulty more precisely. Participant identity was decodable at roughly fifteen times chance from the same session-level embeddings, while medication state was not decodable at all under the same design. Between-participant variation dominates the representation, and the within-participant medication effect is small relative to it. That ordering is the operative cause of the null reported in Section 4.6, and it is a property of the data rather than the pipeline, since the negative control was centred at chance and the identity control cleared its pre-registered ratio criterion. Instability within participants compounds the problem, Van Rheede et al. (2022) attributed roughly 41% of the variance in subthalamic beta power to time of day alone, and Behnke et al. (2025) documented drift in beta peak frequency over the first post-operative months.

One feature family did transfer, weakly. Relative band power gave 0.620 (95% CI [0.519, 0.713]) under the cross-participant task split and 0.629 (95% CI [0.543, 0.714], 24 of 35 participants correctly ordered) under nested selection in the exploratory leave-one-participant-out work. These are the strongest transferable estimates the study produced, and both carry qualifications, the 0.620 figure was not duration-matched and inherits the session-length confound, and the 0.629 figure sits in the pre-registered weak-positive band. That the transferable signal is broadband spectral shape rather than any named oscillation is itself the substantive point, and it is consistent with the argument of Chapter 2 that a method unconstrained by a prior commitment to one band may find information a narrowband analysis would not.

5.5 Spectral dependence and the interpretation of the sweep

Two perturbation results constrain what the highest-scoring arm was reading. Phase randomisation, which preserves the amplitude spectrum while destroying waveform structure,

cost only 0.051 balanced accuracy, and one-second patch occlusion cost between 0.007 and 0.025 with no position-dependent pattern. The model therefore depends on a distributed amplitude-spectrum signature and not on a localised waveform event. The first result also follows from the architecture, since the patch embedding computes the magnitude of a real fast Fourier transform and discards phase. The second is more informative than it appears, a transient account of the result, in which a small number of blinks, swallows or movement events carry the label, predicts a strong position-dependent occlusion effect and is not supported.

The marginal band-stop result requires care rather than interpretation. The most damaging band, 1 to 19 Hz, reduced balanced accuracy from 0.826 to 0.655, but it also removed 98.6% of the power between 1 and 45 Hz, so that figure describes the quantity of power removed rather than the location of the band. Normalising for power removed reverses the emphasis, per unit of power removed, bands between 23 and 41 Hz were 11.8 times more damaging than the sweep average.

That region spans upper beta and low gamma, and it is also where scalp electromyography from cranial and cervical musculature dominates the recorded spectrum. The conditions of this study make a myogenic account substantive. Participants stood, reached and manipulated objects throughout; the electrodes were dry, with higher and less stable impedance than wet electrodes (Hinrichs et al., 2020); the accelerometer triplet was unusable, sitting at approximately 1 g throughout, leaving no independent movement regressor; the automatic component labelling contributed no non-ocular removals in any inspected recording, and its ocular class failed outright; and amplitude rejection was disabled because no threshold retained usable windows without also retaining the artefacts it exists to remove. Chapter 2 is instructive in reverse, Sermon et al. (2023, 2024) and Calvo Peiro et al. (2025) discharged the artefact objection against gamma by direct disconfirming test rather than by assertion, and no equivalent test is available here. The 23 to 41 Hz sensitivity is accordingly reported as uninterpreted, and it should not be advanced as a gamma-band finding.

After the isotonic control within notch width, one position-specific effect survived, an interior optimum centred between 4 and 10 Hz, with a residual of +0.023 at 5 Hz width, present in five of five retraining seeds. This is the closest the study comes to a located spectral dependence, and it is notable that the classical narrow-band sweep on the wider cohort independently placed the largest within-participant OFF-minus-ON elevation between 4.0 and 5.5 Hz, broad and similar in magnitude at lateral and midline channels. Two analyses using different features, analysis sets and validation designs pointed to the same region of the spectrum.

That convergence is a hypothesis rather than a finding, for three reasons. The two estimates come from analysis sets that Chapter 4 does not permit to be compared numerically. At this sample size the individual narrow bands are not statistically distinguishable from one another, and the best narrow band, 2 to 4 Hz at 0.614, did not clear the trivial floor. The fine-tuned model's interior optimum did not reproduce in the reverse task direction. It is nonetheless the most promising lead the study produced, and worth testing directly as an explicit minimal feature under the leak-free certification design.

5.6 The absence of a beta effect

The beta-only classical variants scored 0.486 under both normalisation schemes, which is chance. The a priori hypothesis derived from the most extensively validated marker in this literature was therefore not supported on these recordings, a substantive negative result rather than an omission.

Three accounts are available and none is decisive here. The signature established by Brown et al. (2001) and characterised by Kühn et al. (2006, 2009) is a subcortical one, recorded from within the subthalamic nucleus and the pallidum. Whether an equivalent survives spatial smearing and skull attenuation at the scalp was the open question this dissertation set out to test, and a negative answer is a legitimate one. The cohort was neither implanted nor screened for a beta peak, and the figures of Darcy et al. (2022) and Stanslaski et al. (2024) imply that a substantial minority of hemispheres would lack a usable peak even under invasive recording. The recordings were made during continuous movement, where movement-related beta desynchronisation is large relative to any medication-related difference and is present in both blocks. The band-stop sweep is consistent with the classical result rather than in tension with it, since no beta-specific dependence emerged in the direction that yielded an interpretable interior optimum.

5.7 Methodological implications

Several of the study's controls changed a conclusion. The trivial floor is the clearest case, a classifier given only each session's window count, using no EEG at all, scored 0.614 under the leave-one-participant-out design and exceeded every CodeBrain configuration evaluated there, so without it the 0.557 returned by low-rank adaptation could have been read as a weak positive. The choice of null distribution mattered similarly, the same observed value of 0.657 carrying $p = 0.005$ under a post-hoc permutation test and $p = 0.020$ against a refit-based null whose maximum across 100 repetitions was itself 0.657.

The channel-amplitude control performed both functions a control can perform, falsifying the per-participant task-split design at 0.934 and then confirming the repair at 0.525 under pooling. A design change of that kind is easy to justify by argument and difficult to verify by argument alone. The two withdrawn analyses, the 2 Hz band-stop sweep and all channel-resolved attribution, illustrate the same point from the other direction, both cost reportable results, and neither defect would have been visible to a reader of the output alone.

5.8 Limitations

The fixed pharmacological ordering of the blocks is the most consequential limitation. ON always precedes OFF, so elapsed time, contact drift and fatigue move in the same direction as medication state for every participant, and no design available to this study separates them fully. Labels index the intended state rather than a measured one, and no onset buffer was applied, assessments in both states exist for only 11 of the 35 participants, so restricting the analysis to verified responders was not possible.

The hyperparameter search was run once over the pooled participants rather than inside every outer fold, so selection saw every participant's data and the cross-participant estimates for the adapted arms are optimistic by an unquantified amount. The attribution analysis carries a seed-stability flag, the observed spread of 0.031 crossing the pre-registered flag threshold of ±0.03 without reaching the stop threshold. Attribution was applied to a single arm in a single direction on a single analysis set. The Euclidean alignment experiment specified in the pre-registration was not carried out, that deviation is declared in Section 4.7.

The artefact burden is unquantified in the ways set out in Section 5.5, and 27 participants in the main analysis and 35 in the exploratory work are small numbers against the between-participant variance the results themselves demonstrate.

5.9 Position on the path to clinical use

Chapter 2 distinguished detection from outcome and located almost the entire alternative-biomarker literature at the detection stage. This work sits at that stage and does not leave it. The pre-registered certification gate required a candidate to be rebuilt as a minimal explicit feature and to generalise under a leak-free cross-participant design, and no candidate was certified, relative band power was assessed against the gate and failed it, because its interval does not exclude chance under the required refit-based test. Nothing reported in Chapter 4 supports deploying a model of this kind to a patient it has never seen.

A per-patient calibrated system is a different proposition, and the main analysis bounds what it might achieve on this data. The cost is that calibration requires labelled ON and OFF recordings from each patient, reintroducing the assessment the marker was intended to replace. Its value would lie in continuous monitoring after calibration rather than in removing that burden, and demonstrating it requires across-visit stability evidence a single-visit dataset cannot supply. The one candidate that transferred across participants, relative spectral shape, is weak and is offered as a direction for further work rather than a marker.

6. Conclusion

This dissertation asked whether scalp electroencephalography recorded outside the clinic, with dry electrodes and during free movement, carries enough information to distinguish a patient's ON and OFF levodopa state, and, if so, what frequency content carries it. Chapter 3 documents the work: 35 participants recorded in both medication states while carrying out everyday tasks, a fixed eight-stage preprocessing pipeline, and the CodeBrain foundation model adapted under three arms of increasing capacity. Chapter 4 reports the pooled task split in both directions, its re-run with the test participant withheld, the band-stop attribution sweep, a classical spectral baseline and the earlier leave-one-participant-out work. Chapter 5 sets those results against the literature of Chapter 2 and states the limit of the claim.

The answer has two parts; one settled and one open. Medication state was separable within individuals across a change of behavioural task, at a balanced accuracy of 0.832 under full fine-tuning, with the principal alternative explanations tested and eliminated by the controls of Section 4.3. That separation did not extend to participants whose recordings were absent from training, where every adapted configuration returned to chance and a pairwise test across 702 participant pairs found none that transferred. Two readings remain open, an effect

genuine but individually expressed, or one resting partly on individual recording characteristics that the controls narrow without excluding. The only feature family crossing between individuals was classical relative band power, weakly, at 0.620 under the cross-participant task split and 0.629 under nested selection, beta-band features alone performed at chance. No candidate met the certification criterion fixed in advance.

6.1 Evaluation against the stated objectives

Four objectives were set out in Section 1.5, and all four were met. The first, adapting a pretrained EEG foundation model through parameter-efficient fine-tuning, returned more than asked of it, accuracy rose with every increase in adaptation capacity, which excludes under-adaptation as an explanation for the weaker results and supplies the principal evidence that the successful solution is participant-conditional rather than shared. The second, evaluating that model under a pooled task-split design with both directions reported, produced the study's strongest medication-state estimate and a result the objective did not anticipate, the two directions differ by 0.122 balanced-accuracy points under full fine-tuning against 0.002 under the frozen probe, so reporting both was load bearing rather than a formality.

The third objective, applying perturbation-based explainability to the highest-scoring arm, carried most of the dissertation's weight. It established that the model reads a distributed amplitude-spectrum signature rather than a localised waveform event, that the marginal band-stop profile is dominated by the quantity of power removed rather than its location, and that a single position-specific dependence between 4 and 10 Hz survives a control for that quantity in one direction of the split. It also established what could not be interpreted, since the 23 to 41 Hz sensitivity cannot be separated from muscle contamination with the measurements available and channel-resolved attribution was withdrawn entirely. The fourth objective, establishing the boundary of the result, determined the study's overall conclusion, the controls of Chapter 4 show that the effect is real within participants, is not attributable to recording identity or session length, and did not transfer across individuals at this sample size, for reasons this design cannot identify.

The pre-registration was written against the leave-one-participant-out design, and its predicted outcome, stated in advance as null, held. Its most useful component was not a significance threshold but the trivial floor, a duration-only heuristic at 0.614 that exceeded every CodeBrain configuration evaluated under that design and would otherwise have permitted a weak positive reading of a null result. Two analyses were withdrawn once found to describe the measurement rather than the model, each at the cost of a reportable result. The single declared deviation is recorded in Section 3.8.

6.2 Future work

Five directions follow from the results rather than general aspiration.

The most immediate is to settle the myogenic account of the 23 to 41 Hz sensitivity, which requires measurements this dataset lacks, working accelerometry, or electromyographic channels over the frontalis and temporalis musculature, recorded concurrently so that spectral dependence can be tested against an independent movement regressor.

The second is to test the 4 to 10 Hz candidate directly, rebuilt as a minimal explicit feature and evaluated under the leak-free cross-participant design of Section 3.7. It was not run here, since the fine-tuned model did not reach the design in which it would be meaningful, and remains the shortest route from the present result to a defensible marker.

The third concerns the ordering confound. ON precedes OFF by pharmacological necessity within a visit, but not across visits, so a design recording participants on separate days with the order counterbalanced would separate medication state from elapsed time, contact drift and fatigue, and would supply the across-visit stability evidence any calibrated system requires.

The fourth is per-patient calibration treated as a research question rather than a concession. If the effect is genuine but individually expressed, the operative question is how much labelled data a new patient must supply before performance approaches the within-participant bound reported here, and whether that quantity is clinically tolerable.

The fifth addresses the domain gap. The frozen probe indicates that the pretrained representation does not expose a medication-state axis without substantial adaptation. Continued pretraining on ambulatory dry-electrode data, or the domain-alignment step deferred here, would test whether that gap rather than the signal is the binding constraint.

6.3 Closing remarks

The contribution of this work may not a biomarker, and a study of this size makes it difficult to have delivered one. What it establishes is that levodopa state is decodable from ambulatory dry-electrode EEG within an individual and across a change of task, and that this did not carry to patients the model had not seen. Chapter 2 argued that the field's confidence in its candidate biomarkers rests on evidence that stops short of the clinical question those biomarkers are meant to answer. A study reporting a strong within-participant number without the transfer test that bounds it would have added to that problem. The result is stated at the size the evidence supports, which is less than the headline figure promises and more than a null.

References

Albin, R.L., Young, A.B. and Penney, J.B., 1989. The functional anatomy of basal ganglia disorders. *Trends in neurosciences*, *12*(10), pp.366-375.

Behnke, J.K., Peach, R.L., Habets, J.G., Busch, J.L., Kaplan, J., Roediger, J., Mathiopoulou, V., Feldmann, L.K., Gerster, M., Vivien, J. and Schneider, G.H., 2025. Long-term stability of spatial distribution and peak dynamics of subthalamic beta power in Parkinson's disease patients. *Movement Disorders*, 40(6), pp.1070–1084.

Bronte-Stewart, H.M., Beudel, M., Ostrem, J.L., Little, S., Almeida, L., Ramirez-Zamora, A., Fasano, A., Hassell, T., Mitchell, K.T., Moro, E. and Gostkowski, M., 2025. Long-term personalized adaptive deep brain stimulation in Parkinson disease, a nonrandomized clinical trial. *JAMA Neurology*, 82(11), pp.1171–1180.

Brown, P., Oliviero, A., Mazzone, P., Insola, A., Tonali, P. and Di Lazzaro, V., 2001. Dopamine dependency of oscillations between subthalamic nucleus and pallidum in Parkinson's disease. *Journal of Neuroscience*, 21(3), pp.1033–1038.

Busch, J.L., Kaplan, J., Behnke, J.K., Witzig, V.S., Drescher, L., Habets, J.G. and Kühn, A.A., 2025. Chronic adaptive deep brain stimulation for Parkinson's disease, clinical outcomes and programming strategies. *npj Parkinson's Disease*, 11(1), p.264.

Calvo Peiro, N., Haugland, M.R., Kutuzova, A., Graef, C., Bocum, A., Tai, Y.F., Borovykh, A. and Haar, S., 2025. EEG-based deep learning reveals cortical sensitivity to small changes in deep brain stimulation parameters. *medRxiv*, pp.2025-02.

Cascino, S., Luiso, F., Caffi, L., Palmisano, C., Contaldi, E., Pezzoli, G., Isaias, I.U. and Bonvegna, S., 2026. Chronic adaptive deep brain stimulation in Parkinson's disease, ADAPT-START findings and programming principles. *npj Parkinson's Disease*, 12(1), p.85.

Crook-Rumsey, M., Daniels, S.J., Abulikemu, S., Lai, H., Rapeaux, A., Hadjipanayi, C., Soreq, E., Li, L.M., Bashford, J., Jeyasingh-Jacob, J. and Gruia, D.C., 2023. Multicohort cross-sectional study of cognitive and behavioural digital biomarkers in neurodegeneration, the Living Lab Study protocol. *BMJ open*, *13*(8), p.e072094.

Dadu, A., Satone, V., Kaur, R., Hashemi, S.H., Leonard, H., Iwaki, H., Makarious, M.B., Billingsley, K.J., Bandres-Ciga, S., Sargent, L.J. and Noyce, A.J., 2022. Identification and

prediction of Parkinson's disease subtypes and progression using machine learning in two cohorts. *npj Parkinson's Disease*, 8(1), p.172.

Darcy, N., Lofredi, R., Al-Fatly, B., Neumann, W.J., Hübl, J., Brücke, C., Krause, P., Schneider, G.H. and Kühn, A., 2022. Spectral and spatial distribution of subthalamic beta peak activity in Parkinson's disease patients. *Experimental Neurology*, 356, p.114150.

De Pablo-Fernández, E., Lees, A.J., Holton, J.L. and Warner, T.T., 2019. Prognosis and neuropathologic correlation of clinical subtypes of Parkinson disease. *JAMA Neurology*, 76(4), pp.470–479.

DeLong, M.R., 1990. Primate models of movement disorders of basal ganglia origin. *Trends in neurosciences*, *13*(7), pp.281-285.

Duffley, G., Szabo, A., Lutz, B.J., Mahoney-Rafferty, E.C., Hess, C.W., Ramirez-Zamora, A., Zeilman, P., Foote, K.D., Chiu, S., Pourfar, M.H. and Cnp, C.G., 2023. Interactive mobile application for Parkinson's disease deep brain stimulation (MAP DBS), an open-label, multicenter, randomized, controlled clinical trial. *Parkinsonism & Related Disorders*, 109, p.105346.

Hale, B., Latorre, A., Rocchi, L., Rothwell, J. and Limousin, P., 2026. Clinical utility of evoked potentials for programming subthalamic deep brain stimulation in Parkinson's disease. *npj Parkinson's Disease*.

Hinrichs, H., Scholz, M., Baum, A.K., Kam, J.W., Knight, R.T. and Heinze, H.J., 2020. Comparison between a wireless dry electrode EEG system with a conventional wired wet electrode EEG system for clinical applications. *Scientific reports*, *10*(1), p.5218.

Horváth, K., Aschermann, Z., Ács, P., Deli, G., Janszky, J., Komoly, S., Balázs, É., Takács, K., Karádi, K. and Kovács, N., 2015. Minimal clinically important difference on the Motor Examination part of MDS-UPDRS. *Parkinsonism & Related Disorders*, 21(12), pp.1421–1426.

Kalia, L.V. and Lang, A.E., 2015. Parkinson's disease. *The Lancet*, 386(9996), pp.896–912.

Kühn, A.A., Tsui, A., Aziz, T., Ray, N., Brücke, C., Kupsch, A., Schneider, G.H. and Brown, P., 2009. Pathological synchronisation in the subthalamic nucleus of patients with Parkinson's

disease relates to both bradykinesia and rigidity. *Experimental Neurology*, 215(2), pp.380–387.

Kühn, A.A., Kupsch, A., Schneider, G.H. and Brown, P., 2006. Reduction in subthalamic 8–35 Hz oscillatory activity correlates with clinical improvement in Parkinson's disease. *European Journal of Neuroscience*, 23(7), pp.1956–1960.

Kühn, A.A., Kempf, F., Brücke, C., Gaynor Doyle, L., Martinez-Torres, I., Pogosyan, A., Trottenberg, T., Kupsch, A., Schneider, G.H., Hariz, M.I. and Vandenberghe, W., 2008. High-frequency stimulation of the subthalamic nucleus suppresses oscillatory β activity in patients with Parkinson's disease in parallel with improvement in motor performance. *The Journal of Neuroscience*, 28(24), pp.6165–6173.

Lawhern, V.J., Solon, A.J., Waytowich, N.R., Gordon, S.M., Hung, C.P. and Lance, B.J., 2018. EEGNet, a compact convolutional neural network for EEG-based brain–computer interfaces. *Journal of Neural Engineering*, 15(5), p.056013.

LeCun, Y., Bengio, Y. and Hinton, G., 2015. Deep learning. *Nature*, 521(7553), pp.436–444.

Lofredi, R., Okudzhava, L., Irmen, F., Brücke, C., Huebl, J., Krauss, J.K., Schneider, G.H., Faust, K., Neumann, W.J. and Kühn, A.A., 2023. Subthalamic beta bursts correlate with dopamine-dependent motor symptoms in 106 Parkinson's patients. *npj Parkinson's Disease*, 9(1), p.2.

Louie, K.H., Petrucci, M.N., Grado, L.L., Lu, C., Tuite, P.J., Lamperski, A.G., MacKinnon, C.D., Cooper, S.E. and Netoff, T.I., 2021. Semi-automated approaches to optimize deep brain stimulation parameters in Parkinson's disease. *Journal of NeuroEngineering and Rehabilitation*, 18(1), p.83.

Luo, Y., Qiao, L., Li, M., Wen, X., Zhang, W. and Li, X., 2025. Global, regional, national epidemiology and trends of Parkinson’s disease from 1990 to 2021, findings from the Global Burden of Disease Study 2021. *Frontiers in aging neuroscience*, *16*, p.1498756.

Ma, J., Wu, F., Lin, Q., Xing, Y., Liu, C., Jia, Z. and Feng, M., 2026, April. Codebrain: Bridging decoupled tokenizer and multi-scale architecture for eeg foundation model. In *International Conference on Learning Representations* (Vol. 2026, pp. 83526-83574).

Martinez-Martin, P., Rodriguez-Blazquez, C., Kurtis, M.M., Chaudhuri, K.R. and NMSS Validation Group, 2011. The impact of non-motor symptoms on health-related quality of life of patients with Parkinson's disease. *Movement Disorders*, *26*(3), pp.399-406.

Mathiopoulou, V., Habets, J., Feldmann, L.K., Busch, J.L., Roediger, J., Behnke, J.K., Schneider, G.H., Faust, K. and Kühn, A.A., 2025. Gamma entrainment induced by deep brain stimulation as a biomarker for motor improvement with neuromodulation. *Nature Communications*, 16(1), p.2956.

Merk, T., Peterson, V., Köhler, R., Haufe, S., Richardson, R.M. and Neumann, W.J., 2022a. Machine learning based brain signal decoding for intelligent adaptive deep brain stimulation. *Experimental Neurology*, 351, p.113993.

Merk, T., Peterson, V., Lipski, W.J., Blankertz, B., Turner, R.S., Li, N., Horn, A., Richardson, R.M. and Neumann, W.J., 2022b. Electrocorticography is superior to subthalamic local field potentials for movement decoding in Parkinson's disease. *eLife*, 11, e75126.

Miocinovic, S., de Hemptinne, C., Chen, W., Isbaine, F., Willie, J.T., Ostrem, J.L. and Starr, P.A., 2018. Cortical potentials evoked by subthalamic stimulation demonstrate a short latency hyperdirect pathway in humans. *Journal of Neuroscience*, 38(43), pp.9129–9141.

Neumann, W.J., Degen, K., Schneider, G.H., Brücke, C., Huebl, J., Brown, P. and Kühn, A.A., 2016. Subthalamic synchronized oscillatory activity correlates with motor impairment in patients with Parkinson's disease. *Movement Disorders*, 31(11), pp.1748–1751.

Oehrn, C.R., Cernera, S., Hammer, L.H., Shcherbakova, M., Yao, J., Hahn, A., Wang, S., Ostrem, J.L., Little, S. and Starr, P.A., 2024. Chronic adaptive deep brain stimulation versus conventional stimulation in Parkinson's disease, a blinded randomized feasibility trial. *Nature Medicine*, 30(11), pp.3345–3356.

Ou, Z., Pan, J., Tang, S., Duan, D., Yu, D., Nong, H. and Wang, Z., 2021. Global trends in the incidence, prevalence, and years lived with disability of Parkinson's disease in 204 countries/territories from 1990 to 2019. *Frontiers in public health*, *9*, p.776847.

Pedrosa de Menezes, A.L., Bloem, B.R., Beckers, M., Piat, C., Benarroch, E.E. and Savica, R., 2024. Molecular variability in Levodopa absorption and clinical implications for the management of Parkinson’s disease. *Journal of Parkinson’s Disease*, *14*(7), pp.1353-1368.

Poewe, W., Seppi, K., Tanner, C.M., Halliday, G.M., Brundin, P., Volkmann, J., Schrag, A.E. and Lang, A.E., 2017. Parkinson disease. *Nature Reviews Disease Primers*, 3(1), pp.1–21.

Postuma, R.B., Berg, D., Stern, M., Poewe, W., Olanow, C.W., Oertel, W., Obeso, J., Marek, K., Litvan, I., Lang, A.E. and Halliday, G., 2015. MDS clinical diagnostic criteria for Parkinson's disease. *Movement disorders*, *30*(12), pp.1591-1601.

Roy, Y., Banville, H., Albuquerque, I., Gramfort, A., Falk, T.H. and Faubert, J., 2019. Deep learning-based electroencephalography analysis, a systematic review. *Journal of Neural Engineering*, 16(5), p.051001.

Schirrmeister, R.T., Springenberg, J.T., Fiederer, L.D.J., Glasstetter, M., Eggensperger, K., Tangermann, M., Hutter, F., Burgard, W. and Ball, T., 2017. Deep learning with convolutional neural networks for EEG decoding and visualization. *Human Brain Mapping*, 38(11), pp.5391–5420.

Sermon, J.J., Olaru, M., Anso, J., Cernera, S., Little, S., Shcherbakova, M., Bogacz, R., Starr, P.A., Denison, T. and Duchet, B., 2023. Sub-harmonic entrainment of cortical gamma oscillations to deep brain stimulation in Parkinson's disease, model based predictions and validation in three human subjects. *Brain Stimulation*, 16(5), pp.1412–1424.

Sermon, J.J., Benjaber, M., Duchet, B., Anso, J., Olaru, M., Starr, P.A. and Denison, T., 2024. 1:2 entrainment is not a device-induced artefact, except when it is. *Brain Stimulation*, 17(2), p.149.

Siderowf, A., McDermott, M., Kieburtz, K., Blindauer, K., Plumb, S., Shoulson, I. and Parkinson Study Group, 2002. Test–retest reliability of the unified Parkinson's disease rating scale in patients with early Parkinson's disease, results from a multicenter clinical trial. *Movement Disorders*, 17(4), pp.758–763.

Stanslaski, S., Summers, R.L., Tonder, L., Tan, Y., Case, M., Raike, R.S., Morelli, N., Herrington, T.M., Beudel, M., Ostrem, J.L. and Little, S., 2024. Sensing data and methodology from the Adaptive DBS Algorithm for Personalized Therapy in Parkinson's Disease (ADAPT-PD) clinical trial. *npj Parkinson's Disease*, 10(1), p.174.

Sun, L. and Hinrichs, H., 2016. Moving average template subtraction to remove stimulation artefacts in EEGs and LFPs recorded during deep brain stimulation. *Journal of neuroscience methods*, *266*, pp.126-136.

Tanimura, J., Yako, T. and Hashimoto, T., 2025. Chronic adaptive versus conventional deep brain stimulation in patients with Parkinson's disease, a randomized crossover trial with exploratory analysis of baseline-dependent response patterns. *medRxiv*, pp.2025-09.

Tinkhauser, G., Pogosyan, A., Tan, H., Herz, D.M., Kühn, A.A. and Brown, P., 2017a. Beta burst dynamics in Parkinson's disease OFF and ON dopaminergic medication. *Brain*, 140(11), pp.2968–2981.

Tinkhauser, G., Pogosyan, A., Little, S., Beudel, M., Herz, D.M., Tan, H. and Brown, P., 2017b. The modulatory effect of adaptive deep brain stimulation on beta bursts in Parkinson's disease. *Brain*, 140(4), pp.1053–1067.

van Rheede, J.J., Feldmann, L.K., Busch, J.L., Fleming, J.E., Mathiopoulou, V., Denison, T., Sharott, A. and Kühn, A.A., 2022. Diurnal modulation of subthalamic beta oscillatory power in Parkinson's disease patients during deep brain stimulation. *npj Parkinson's Disease*, 8(1), p.88.

Walker, H.C., Huang, H., Gonzalez, C.L., Bryant, J.E., Killen, J., Cutter, G.R., Knowlton, R.C., Montgomery, E.B., Guthrie, B.L. and Watts, R.L., 2012. Short latency activation of cortex during clinically effective subthalamic deep brain stimulation for Parkinson's disease. *Movement Disorders*, 27(7), pp.864–873.

Wang, J., Zhao, S., Luo, Z., Zhou, Y., Jiang, H., Li, S., Li, T. and Pan, G., 2025, May. Cbramod, A criss-cross brain foundation model for eeg decoding. In *International conference on learning representations* (Vol. 2025, pp. 75310-75346).

Wiest, C., Tinkhauser, G., Pogosyan, A., He, S., Baig, F., Morgante, F., Mostofi, A., Pereira, E.A., Tan, H., Brown, P. and Torrecillos, F., 2021. Subthalamic deep brain stimulation induces finely-tuned gamma oscillations in the absence of levodopa. *Neurobiology of Disease*, 152, p.105287.

World Health Organization, 2022. *Parkinson disease, A public health approach. Technical brief*. World Health Organization.

Appendix A

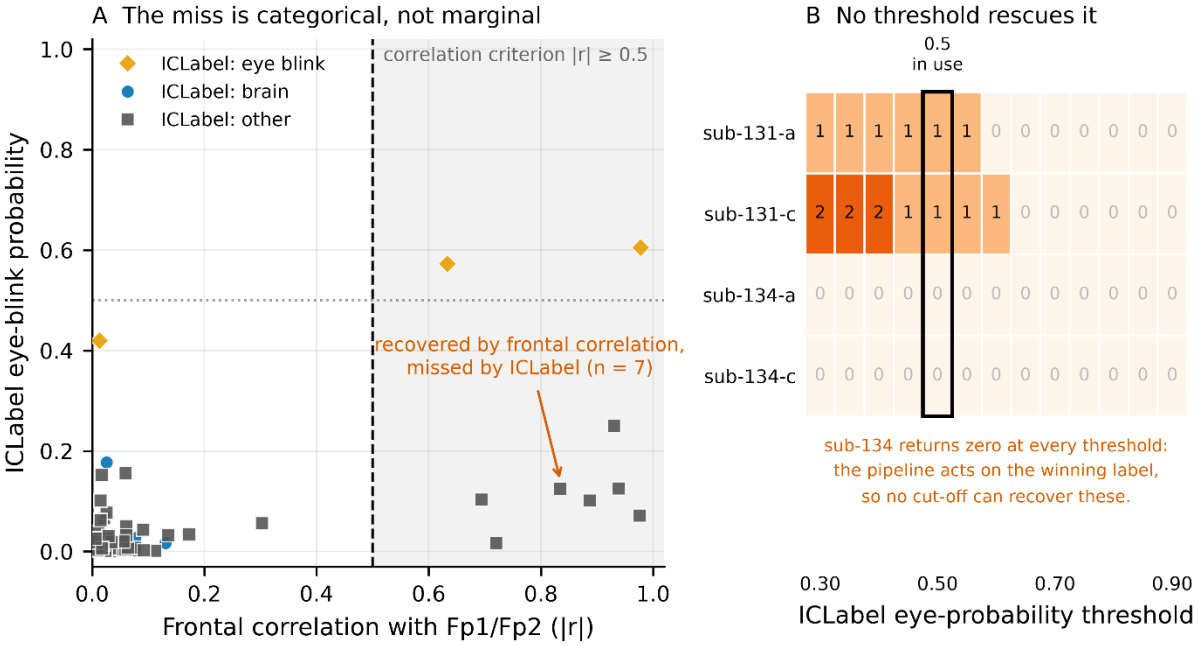


Figure A-1. The two ICLabel detector behaviours that make the 0.5 threshold unsafe to transfer. (A) Every independent component from the four inspected recordings, plotted by its correlation with the frontal channels Fp1 and Fp2 against the eye-blink probability ICLabel assigned it. The shaded region is the correlation criterion used. Nine components exceed that criterion and ICLabel classes only three of them as eye blinks; the seven it misses carry eye probabilities between 0.017 and 0.250 despite frontal correlations as high as 0.976. They are not borderline cases that a slightly lower cut-off would have caught, but components confidently assigned to the wrong class. (B) The number of components ICLabel classes as eye blinks in each recording, swept across the full range of probability thresholds. Because the pipeline acts on the winning label rather than on the probability, lowering the threshold cannot recover a component whose winning label is something else, both sessions of sub-134 return zero at every threshold from 0.30 to 0.90, while sub-131 returns one or two. The same threshold, applied to the same montage under the same pipeline, therefore, behaves quite differently in two recordings from the same cohort, which is why the value chosen here should not be carried to a new recording without being re-checked.

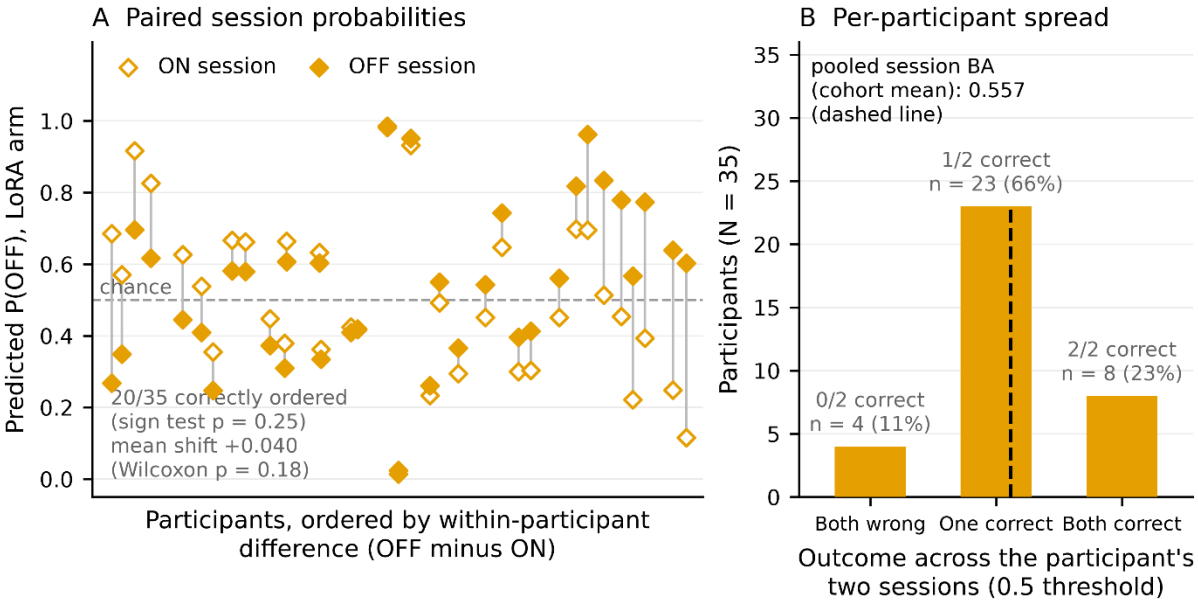


Figure A-2. Paired session probabilities and per-participant spread for the primary LoRA arm (exploratory LOPO design, N = 35 participants). (A) Each participant's ON-session and OFF-session predicted P(OFF), joined by a line, ordered by within-participant difference (OFF minus ON): 20 of 35 participants correctly ordered (one-sided sign test p = 0.25), mean shift +0.040 (one-sided Wilcoxon signed-rank p = 0.18). (B) The same arm's outcome per participant when each session is separately thresholded at 0.5, which can only take three values with one ON and one OFF session each; the pooled cohort mean (0.557, Table 4-2) is marked for contrast. Panel A is a threshold-free ordering test; panel B applies the 0.5 decision threshold to each session independently, so the two panels' counts do not reconcile. A participant can be correctly ordered in A while one of their two sessions still falls on the wrong side of 0.5 in B.

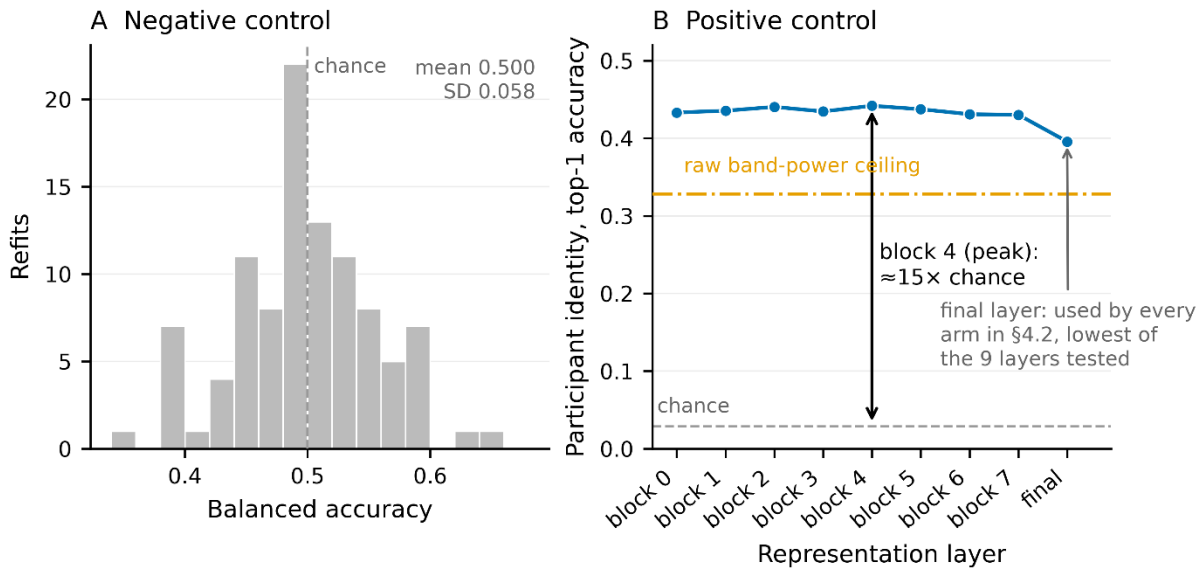


Figure A-3. Negative and positive controls for the exploratory LOPO design (N = 35 participants, 70 sessions). (A) Negative control, the null distribution of balanced accuracy over 100 complete refits of the pipeline with medication labels shuffled within each participant, centred at 0.500 (SD 0.058). No refit exceeded 0.657, the classical candidate's observed value (Section 4.6), giving that value a refit-based p of 0.020. (B) Positive control, participant-identity decoding (35-way, chance 0.029) from CodeBrain's representation at each of its eight residual blocks and at the final layer, against the ceiling supported by the raw band-power input (0.328). Every layer clears the ceiling; the peak, block 4, decodes identity at approximately 15 times chance. The final layer, the one every arm in Section 4.2 uses, is the weakest of the nine. Points in (B) are single-split estimates with no per-layer interval; the curve's smoothness should not be read as measured precision.

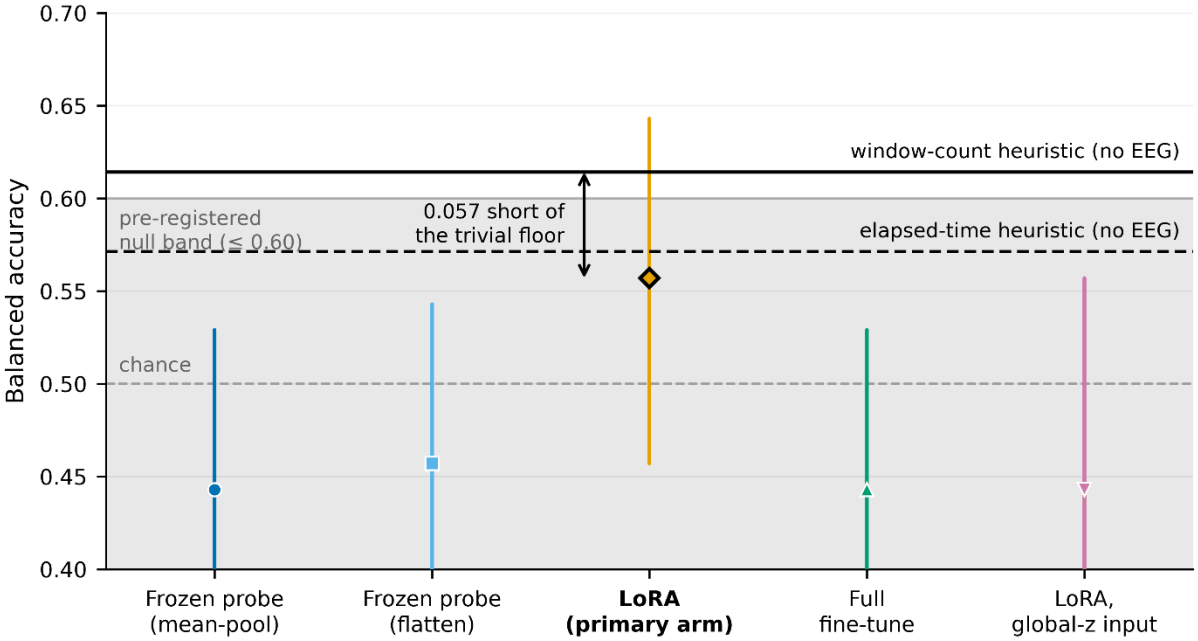


Figure A-4. The trivial floor, two no-EEG heuristics against the five CodeBrain configurations (exploratory LOPO design, N = 35 participants, 70 sessions). Balanced accuracy with 95% cluster-bootstrap confidence intervals resampled over participants (2,000 resamples). The shaded region marks the pre-registered null band (≤ 0.60). The window-count heuristic (0.614) and the elapsed-time heuristic (0.571) use no EEG signal at all; neither heuristic has a stored interval, so both are drawn as reference lines rather than as points with a fabricated error bar. LoRA, the pre-registered primary arm (bold, black-edged marker), is the highest-scoring configuration and still falls 0.057 short of the window-count heuristic.

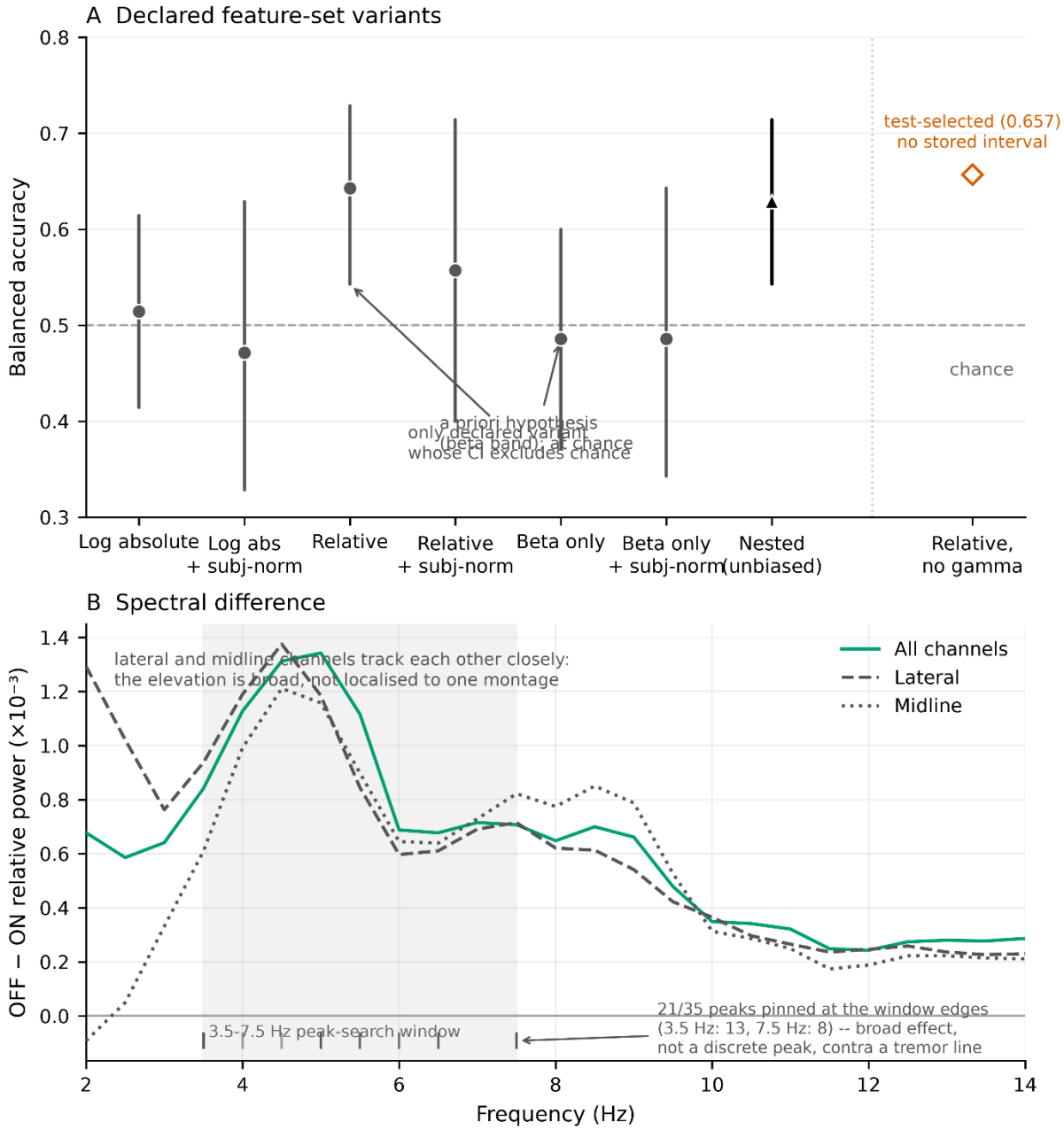


Figure A-5. The classical spectral candidate (exploratory LOPO design, N = 35 participants, 70 sessions). (A) The six pre-declared relative/log-absolute/beta feature-set variants (with and without per-subject normalisation), each with 95% cluster-bootstrap confidence intervals (2,000 resamples); the nested, selection-unbiased estimate (0.629); and the test-selected variant, relative power without gamma (0.657), separated by a rule because only its point estimate is stored, with no interval. Relative power (undeclared subject-norm, all bands) is the only declared variant whose interval excludes chance. The a priori hypothesis, the beta band, sits at chance. (B) Within-participant OFF-minus-ON relative power across 0.5 Hz bins, 2 to 14 Hz (Welch, 2 s segments, 0.5 Hz resolution), split by lateral and midline channels, which track each other closely. The rug marks each

participant's own peak frequency, located by an argmax restricted to a fixed 3.5-7.5 Hz search window: 21 of 35 peaks land exactly on one of the two window edges (13 at 3.5 Hz, 8 at 7.5 Hz), which is the signature of a broad effect the window is too narrow to bound and is evidence against a discrete tremor-frequency peak, not for one.